

CGVT-FSL: Concept-guided visual-textual few-shot learning for cross-domain hyperspectral image classification[☆]

Haojin Tang^a , Jiaqing Peng^a, Hongyi Li^b , Xiaofei Yang^{a,*}, Dong Tang^a, Xin Lin^c, Weixin Xie^d

^a School of Electronic and Communication Engineering, Guangzhou University, Guangzhou, 511370, China

^b School of Science and Engineering, Chinese University of Hong Kong, Shenzhen, 518172, China

^c College of Artificial Intelligence, Guangzhou University, Guangzhou, 511370, China

^d Guangdong Provincial Key Laboratory of Intelligent Information Processing, Shenzhen University, Shenzhen, 518060, China

HIGHLIGHTS

- TAFSL integrates domain-agnostic text with visual features for cross-domain few-shot HSI classification.
- Domain-Agnostic Class-Label Text Description generates class-label descriptions capturing intra-class similarity and inter-class differences.
- VTDFormer jointly extracts visual and textual spatial-spectral features across source and target domains.
- SSTDAM uses textual semantics to align multimodal feature distributions and reduce domain shift.
- Experiments on six HSI datasets show superior performance over state-of-the-art few-shot methods.

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ABSTRACT

Visual-textual few-shot learning (VT-FSL) models have recently shown great potential in cross-domain hyperspectral image (HSI) classification, as textual descriptions provide domain-invariant semantic knowledge that enhances visual generalization. However, existing VT-FSL methods typically perform global image-label alignment using simple class names or short label phrases, capturing only coarse-grained semantics and lacking explicit conceptual cues aligned with visual spatial-spectral features. Consequently, the resulting visual-textual correspondence remains sensitive to domain shifts. To address this issue, we argue that concept-level semantics such as material or density are more stable and transferable across domains. Therefore, we propose a novel concept-guided visual-textual few-shot learning (CGVT-FSL) framework. First, we design a concept-embedded class-label text description (C2TD) method that incorporates LLM-generated concept knowledge into class-level textual representations, enabling interpretable and transferable semantic alignment. Second, we develop a visual-textual dual-branch transformer (VTDFormer), which jointly extracts visual spatial-spectral features and concept-guided textual features across domains. Moreover, a spatial-spectral-textual domain adversarial module (SSTDAM) is introduced to further mitigate domain shift by leveraging the shared concept-embedded linguistic domain. Extensive experiments on six HSI datasets demonstrate that CGVT-FSL consistently outperforms state-of-the-art FSL methods. The codes are available from the website: <https://github.com/HongyiLi-GZHU/CGVT-FSL>.

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* Corresponding author.

Email addresses: tanghaojin@gzhu.edu.cn (H. Tang), pjq@e.gzhu.edu.cn (J. Peng), 225010340@link.cuhk.edu.cn (H. Li), xiaofeiayang@gzhu.edu.cn (X. Yang), tangdong@gzhu.edu.cn (D. Tang), linxin94@gzhu.edu.cn (X. Lin), wxxie@szu.edu.cn (W. Xie).

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1. Introduction

Hyperspectral images (HSIs) stand out for their ability to provide rich spectral information about land-cover classes, which enables highly accurate recognition of ground objects. Nowadays, HSIs have been widely applied in many fields, such as environmental monitoring, resource management, military reconnaissance and city planning [1,2]. HSI classification is a key technology in HSI processing, which assigns pixels in the HSI to specific classes [3,4]. However, the complexity and risk of the HSI data collection environment pose a small-sample-size challenge for HSI classification. Recent studies show that cross-domain information can help tackle the problem of limited labeled samples. Hence, cross-domain HSI classification has gained significant attention as a major research topic [5–8]. In cross-domain HSI classification, the source domain contains sufficient labeled samples, while the target domain has fewer labeled samples. Such a cross-domain task setting naturally leads to a domain shift, which stems from combined differences in geographic location, spectral band range, and sensor type, among other factors. Moreover, a key challenge in this setting is the presence of “unseen classes” in the target domain, which refer to categories absent from the source domain. This is a defining characteristic that distinguishes cross-domain from cross-scene problems, as the latter assumes no unseen classes across domains [9]. Therefore, the core idea of cross-domain HSI classification is to train a model by learning transferable knowledge from the source domain, which can be effectively generalized to the target domain with only a few labeled samples.

Few-shot learning (FSL) is a prominent approach in cross-domain learning, enabling the development of generalized models through the construction of cross-domain episodes [10,11]. Generally, FSL methods are typically categorized into two types, including meta-learning-based methods [12] and metric-learning-based methods [13]. In terms of HSI processing, current research mainly employs the metric-learning-based methods for cross-domain HSI classification. For instance, deep few shot learning (DFSL) [14] and deep cross-domain few shot learning (DCFSL) [15] are classic FSL applications in cross-domain HSI classification. Both methods learn a metric space from the source domain and use it to classify HSI pixels in the target domain. Notably, DCFSL, inspired by CDAN [16], introduces domain adversarial learning to explicitly align cross-domain feature distributions, achieving significant success and advancing the use of domain adversarial learning to mitigate domain shift. Although FSL and domain adversarial learning methods have made significant improvements in cross-domain HSI classification, most FSL methods focus on learning domain-invariant representations at the visual level, ignoring the importance of textual information.

Text information provides valuable prior knowledge about ground objects, which aids in achieving object recognition and classification. Moreover, this type of information is generally independent of specific data domains. With the rapid development of deep learning and natural language processing technologies, multi-modal learning-based models [17], particularly image-text models, have demonstrated excellent performance in various image processing tasks [18–20], including FSL tasks. For instance, the contrastive language-image pre-training model called CLIP [21] has shown remarkable performance in both few-shot and zero-shot learning tasks. Recently, the linguistic modality has been explored in cross-domain and cross-scene HSI classification. For example, the language-aware domain generalization network (LDGnet) [9] has been successfully applied to cross-scene HSI classification. More recently, Li et al. [22] proposed a robust few-shot HSI classification framework with cross-modal alignment and supervised contrastive learning. Li et al. [8] proposed Semantic Guided Prototype Learning for Cross-Domain Few-shot Hyperspectral Image Classification model (SGPL) which can integrate semantic information of class labels into the prototype learning framework, thereby enhancing the expressive capacity of each category prototype. Despite these advances, the application of visual-textual FSL (VT-FSL) in cross-domain HSI classification is still in its early stages. Most existing approaches align image representations

with class names or short label phrases, which only capture coarse-grained semantics. Such label-level textual descriptions lack explicit concept-level cues that are more stable and transferable across domains. Therefore, existing VT-FSL methods have not fully leveraged the adversarial learning framework by treating text as an independent domain, separate from the source and target domains, to address the gap in cross-domain. Consequently, the visual-textual correspondence established by these models is easily affected by domain shifts.

While seminal multimodal HSI classification models primarily address single-domain or cross-scene tasks, existing approaches for cross-domain scenarios suffer from three key limitations:

1. Current class-label text description methods are either overly simplistic, relying only on plain class names, or too rigid to generalize to unseen categories, and they rarely incorporate transferable concept-level semantics.
2. Existing cross-domain HSI classification methods fail to jointly model visual spatial-spectral features and semantically informative textual representations, and often rely on coarse class-level semantics, thereby limiting fine-grained visual-textual alignment and weakening semantic transfer across domains.
3. The potential of the linguistic modality as a domain-invariant bridge is underexplored in domain adversarial learning frameworks for cross-domain HSI classification.

To address the aforementioned challenges, we propose the concept-guided visual-textual few-shot learning (CGVT-FSL) framework, a novel multimodal approach that leverages concept-level linguistic information to enhance cross-domain HSI classification. As shown in Fig. 1, we hypothesize that concept-embedded semantics provide a shared and domain-invariant linguistic space that bridges the gap between source and target visual domains. Unlike conventional label-level textual representations that rely solely on class names, the concept-embedded textual descriptions encode fine-grained attributes such as material and density, which remain consistent across domains. These concept-embedded descriptions also capture the relationships among all samples within a category and the semantic distinctions and connections between different categories (e.g., both “grass” and “trees” share the high-level concept of “Natural”), which are known as intra-class and inter-class relationships. By aligning visual spectral-spatial features with these concept-guided textual embeddings through a dual-branch architecture and cross-modal adversarial learning, CGVT-FSL enables more interpretable and transferable domain adaptation. The main contributions of this work are summarized as follows:

1. We propose a concept-guided text construction strategy, termed concept-embedded class-label text description (C2TD), which integrates LLM-generated concept knowledge into class-level textual representations. By converting class labels into structured concept-embedded textual descriptions, C2TD captures both intra-class attributes and inter-class relationships, thereby providing domain-agnostic and interpretable semantic cues as inputs to the subsequent visual-textual learning framework and improving generalization to unseen categories.
2. We propose a visual-textual dual-branch transformer (VTDFormer) to address the lack of unified modeling of visual spatial-spectral features and semantically informative textual representations in existing cross-domain HSI classification methods. It jointly learns visual spatial-spectral features and concept-guided textual representations, thereby establishing fine-grained visual-textual correspondence through self-attention and a visual-linguistic-prototype alignment strategy.
3. We propose a spatial-spectral-textual domain adversarial module (SSTDAM) to further explore the underexplored role of the linguistic modality in domain adaptation. By leveraging text as a domain-invariant semantic bridge and conducting both

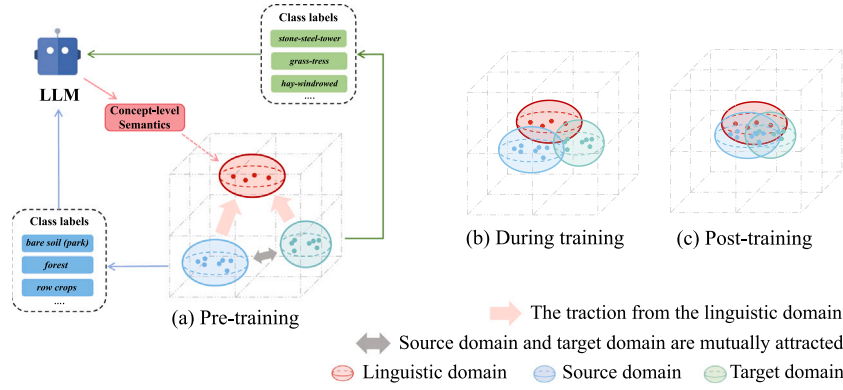


Fig. 1. The evolution of visual-textual alignment across the linguistic, source, and target domains during training. The proposed CGVT-FSL method progressively bridges the semantic gap by aligning visual representations with concept-guided linguistic semantics.

visual-linguistic and source-target adversarial learning, SSTDA mitigates the residual source-target distribution discrepancy.

- Comprehensive experiments on six benchmark HSI datasets demonstrate that the proposed CGVT-FSL achieves superior performance compared to state-of-the-art FSL methods, validating the effectiveness of concept-guided cross-modal alignment in addressing domain shifts.

2. Related work

2.1. Visual-textual few-shot learning

In VT-FSL, the design of textual descriptions is critically important, as they provide essential semantic guidance for learning visual representations and directly impact model generalization in few-shot scenarios. This has motivated extensive research on improving textual prompts for vision-language learning. For instance, Liu et al. [23] proposed RemoteCLIP based on standard CLIP templates, while Roth et al. proposed WaffleCLIP [24], which leverages large language models (LLMs) to generate descriptive text and alleviate class name ambiguity. Cheng et al. [25] further proposed TeAw to construct detailed contextual descriptions of land-cover classes, and LDGNet [9] incorporates spatial context into textual prompts for cross-scene classification. Beyond single-modal prompt optimization, recent advances in vision-language models have highlighted the limitations of uni-modal prompting strategies. Khattak et al. [26] proposed MaPLe, which jointly adapts visual and textual prompts and demonstrates that explicit cross-modal coupling is critical for improving out-of-distribution generalization. More recently, in-context learning has been explored for adapting vision and vision-language models under distribution shifts. For example, Zhou et al. [27] analyze the role of in-context learning in multimodal adaptation, while Xie et al. [28] propose test-time visual in-context tuning to improve robustness on unseen domains. However, these methods are primarily developed for natural-image settings and do not explicitly address the challenges of cross-domain hyperspectral few-shot classification, where semantic transfer across domains, limited labeled samples, and spectral-spatial heterogeneity must be jointly considered.

Despite these advances, existing VT-FSL methods still exhibit limited cross-domain generalization capability. Specifically, current approaches mainly rely on coarse class-level textual semantics and lack transferable concept-level knowledge that can bridge domain gaps. Moreover, most methods depend on relatively shallow linguistic priors, which restrict the ability to establish fine-grained visual-textual alignment and weaken semantic transfer under domain shifts. These limitations motivate the need for concept-guided semantic modeling and deeper cross-modal interaction for robust cross-domain HSI classification.

2.2. Cross-domain HSI classification

HSI classification is a crucial technology in the field of HSI processing, with the objective of categorizing all pixels within the HSI into specific classes [29–33]. Nowadays, deep learning-based methods have become the mainstream techniques in HSI classification [34–36]. Li et al. [37] designed a 3D convolutional neural network (3D-CNN) framework that significantly improves HSI classification performance, achieving remarkable results. To fully combine spatial and spectral information, Zhong et al. [38] proposed an end-to-end spectral-spatial residual network (SSRN) to address the spatial-spectral feature extraction problem, which exhibits strong classification performance with both small and large training sample sizes. Sun et al. [39] proposed a memory-augmented spectral-spatial transformer framework that combines the strengths of CNNs and transformers, thereby improving the robustness of spatial-spectral feature representation. However, deep learning-based methods require large amounts of labeled data, which pose significant challenges due to the scarcity of labeled HSI samples.

Recent research suggests that cross-domain information can effectively address this challenge by leveraging labeled samples from different domains. Consequently, cross-domain HSI classification [40] has garnered significant attention, emerging as a prominent research focus. Liu et al. [14] pioneered the application of FSL to cross-domain HSI classification with their DFSL method, which learns a generalized metric space for classifying target domain samples. To integrate with the domain adaptation strategy, Li et al. [15] designed a novel FSL framework called DCFSL for cross-domain HSI classification, which effectively aligns features between source and target domains. This work has significantly advanced the development of FSL frameworks in cross-domain HSI classification. As an improvement to DCFSL, Huang et al. [41] introduced an improved spatial-spectral transformer architecture for hyperspectral few-shot classification (HFC-SST). This method explores the transformer architecture to extract local spatial-spectral information, improving generalization capability and boosting cross-domain HSI classification accuracy. Furthermore, Zhang et al. [42] presented a graph information aggregation cross-domain few-shot learning (GiacFSL) framework for HSI classification, which aims to exploit nonlocal relationships to mitigate the effects of domain shift. With respect to state-of-the-art FSL methods for cross-domain HSI classification, Liu et al. [43] designed a multi-level relation learning (MRLF) method for cross-domain HSI classification. This method emphasizes the use of contrastive learning to distinguish relationships among class-level samples, enabling the extraction of more distinctive spatial-spectral features. Liu et al. [44] proposed a hybrid multitask learning (HyMuT) framework for cross-domain HSI classification, which integrates different tasks to extract domain-invariant spatial-spectral features. This method establishes a unified network architecture with self-supervised

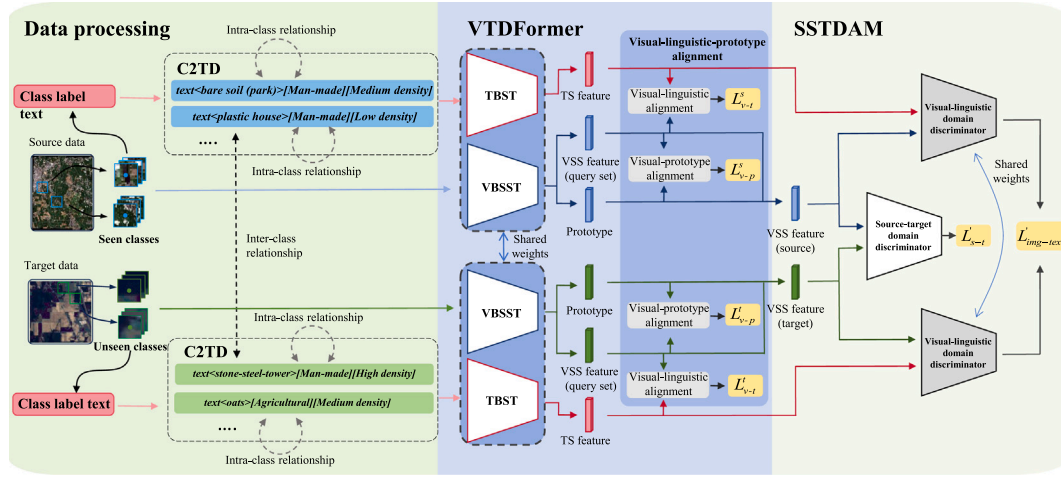


Fig. 2. Overview of the proposed CGVT-FSL framework. C2TD denotes a concept-guided text construction strategy that transforms class labels into structured concept-embedded textual descriptions, providing domain-agnostic semantic inputs for the textual branch. VTDFormer jointly processes the visual data and the concept-guided textual representations to learn unified cross-modal features. A visual-linguistic prototype alignment strategy is then applied to enhance the consistency between visual and textual representations. Finally, SSTDAM performs domain-adversarial learning by leveraging textual semantics as a domain-invariant bridge, thereby reducing source-target discrepancy and improving cross-domain generalization.

learning to exploit domain and task invariance, demonstrating superior classification results compared to advanced methods. Despite these advancements, existing methods fail to incorporate transferable concept-level semantics or effectively utilize concept-guided textual knowledge to enhance the domain-generalization capability of models.

3. Proposed method

As illustrated in Fig. 2, we first apply C2TD as a concept-guided text construction strategy to convert class labels into structured concept-embedded textual descriptions. These descriptions provide interpretable and domain-agnostic semantic cues that encode both intra-class attributes and inter-class relationships, and serve as semantic inputs to the textual branch. Based on this, the proposed VTDFormer jointly learns visual spatial-spectral features and concept-guided textual representations, enabling unified cross-modal modeling and enhancing the discriminative structure of the shared feature space. A visual-linguistic prototype alignment strategy is further introduced to align visual representations with textual semantics, facilitating robust knowledge transfer across domains. Finally, the SSTDAM module performs domain-adversarial learning to reduce source-target discrepancy by leveraging textual semantics as a domain-invariant bridge, thereby improving cross-domain generalization.

3.1. Concept-embedded class-label text description (C2TD)

In previous visual-textual models for HSI [9], class names are typically used as the main textual cues for semantic representation at each HSI pixel. These models generally adopt a simple template such as “A hyperspectral image of <class name>”, and occasionally extend it with additional scene-dependent spatial context, e.g., “The stressed grass is next to the road.” However, such descriptions primarily rely on class labels and overlook the concept-level semantics (e.g., material and density) that are more transferable and domain-invariant. As a result, the generated textual features are often coarse-grained and lack the conceptual grounding necessary for robust visual-textual alignment across domains.

To overcome this limitation, we introduce C2TD as a concept-guided text construction strategy rather than a standalone parameterized feature-extraction module. C2TD transforms class labels into structured textual descriptions enriched with transferable concept-level semantics,

thereby providing the textual branch with more domain-agnostic and semantically informative inputs. Specifically, we first collect all class names from both the source and target domains. As shown in Fig. 3, these class names are then provided to a large language model (LLM) using a unified prompt template. We employ three concept categories to generate conceptual attributes: [Natural], [Man-made], and [Agricultural] for object types; [High density], [Medium density], and [Low density] for material density; and [High specific heat capacity], [Medium specific heat capacity], and [Low specific heat capacity] for thermal properties. The prompt used is: “I will provide the classification names for the corresponding pixel points of some hyperspectral images. Please categorize them into the following three types based on their physical properties and your existing knowledge: [Natural], [Man-made], and [Agricultural]. Here are the classification names: {all class names from source domain and target domain}.” Similar prompts are used for the other concept categories. Since the reliability of LLM-generated concept labels is not uniform across different semantic dimensions, we further introduce a manual review step to ensure the correctness and consistency of the class-to-concept label matching. Following the framework for human evaluation of LLM outputs proposed in [45], we explicitly define the manual review procedure. The review is conducted at the class level under the predefined C2TD taxonomy across three concept dimensions, namely object type, density, and specific heat capacity. For each land-cover class, the LLM-generated concept labels are examined by authors with remote-sensing expertise according to fixed class-level criteria derived from (i) the semantic meaning of the class name, (ii) general remote-sensing knowledge of land-cover properties, and (iii) the examples and figure annotations provided in this manuscript. The reviewed labels are then used as reference annotations for subsequent evaluation. After this review step, the verified concept labels are embedded into structured text templates and used as semantic inputs for the subsequent visual-textual learning framework. For example, the class “grass” can be represented as “A hyperspectral image of grass [Natural][Medium density][Low specific heat capacity].”

This concept embedding process effectively bridges the semantic gap between seen and unseen classes by encoding domain-agnostic conceptual knowledge within the textual modality. Consequently, C2TD provides a richer and more interpretable representation that captures both intra-class consistency and inter-class distinctiveness, forming a robust foundation for visual-textual semantic alignment and domain-adversarial learning in the subsequent modules.

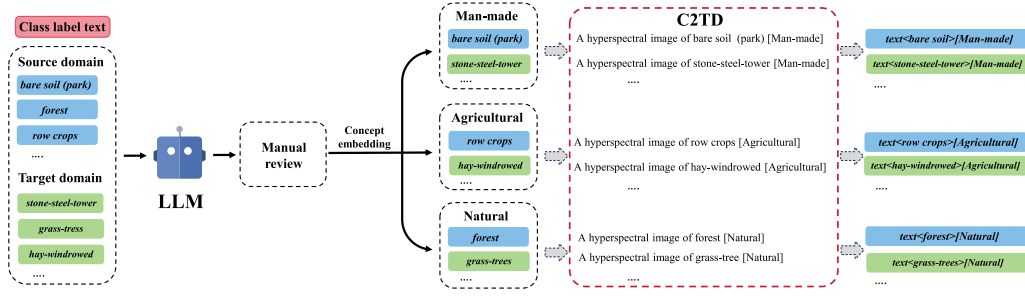


Fig. 3. Illustration of the proposed C2TD method.

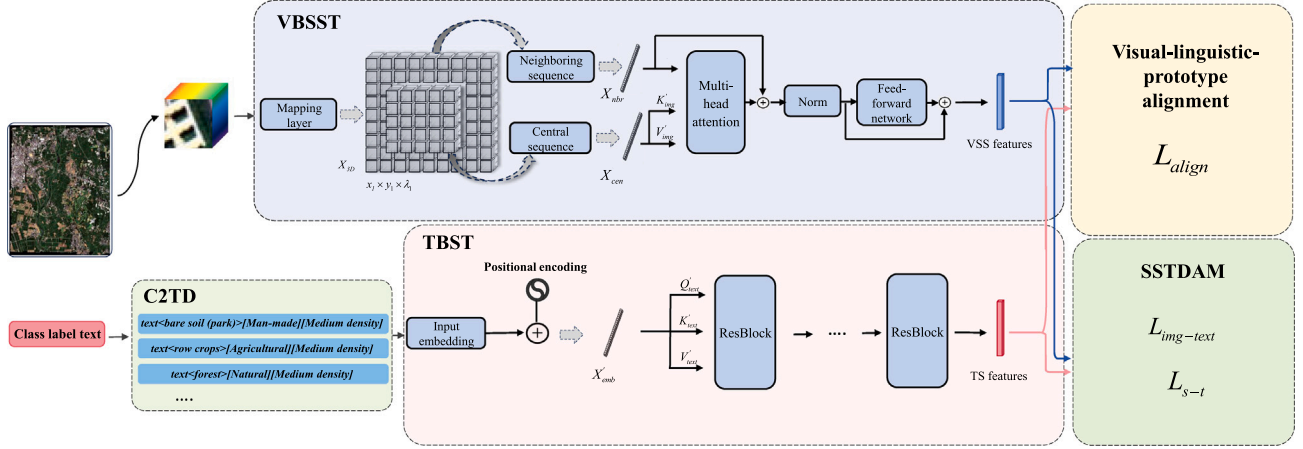


Fig. 4. Illustrative framework of the proposed VTDFormer.

3.2. Visual-textual dual-branch transformer (VTDFormer)

In the area of computer vision, CNN-based methods [46,47] have been widely used for HSI feature extraction, benefiting from local receptive fields and parameter sharing. However, their limited receptive range restricts the ability to model long-range spatial-spectral dependencies. In contrast, transformers [48,49] have demonstrated remarkable performance in both natural language and vision tasks by capturing global contextual relationships through self-attention. Despite this progress, a common limitation persists: many image-text foundation models, including early and influential variants of CLIP, rely on a hybrid architecture (e.g., employing CNNs for visual features and transformers for textual features). This design prevents unified modeling of cross-modal global context and can limit the model's ability to capture consistent semantics across modalities and domains.

To address these limitations, we propose a dual-branch transformer architecture, termed VTDFormer, for cross-domain HSI classification. VTDFormer jointly models spatial-spectral dependencies in the visual modality and semantic dependencies in the textual modality within a unified framework, thereby enhancing fine-grained visual-textual alignment and facilitating subsequent domain adaptation.

The overall architecture of VTDFormer is illustrated in Fig. 4. It consists of two complementary components: the visual-branch spatial-spectral transformer (VBSST) encoder and the textual-branch semantic transformer (TBST) encoder. These two branches adopt distinct designs to accommodate the intrinsic characteristics of their respective modalities. Specifically, VBSST processes three-dimensional hyperspectral patches with strong local spatial-spectral correlations, and incorporates a local center-neighborhood attention mechanism to capture fine-grained discriminative patterns. In contrast, TBST processes textual sequences dominated by semantic dependencies and employs standard Transformer self-attention to model long-range contextual relationships.

This branch is designed to learn concept-guided semantic representations for subsequent visual-textual alignment, rather than encoding spatial-spectral properties in the textual modality. This modality-aware design enables each branch to effectively model its corresponding representation while jointly contributing to a unified visual-textual feature space. In the VBSST encoder, three-dimensional patches are first extracted from the HSI to represent individual pixels, following the strategy in [41]. These patches are then partitioned into a central sequence X_{cen} and a neighboring sequence X_{nbr} to facilitate localized self-attention-based modeling. Specifically, let X_{3D} denote the 3-D patch associated with a given HSI pixel. The segmentation function $\varphi(\cdot)$ applied to X_{3D} is formulated as follows:

$$[X_{cen}, X_{nbr}] = \varphi(X_{3D}, \omega) \quad (1)$$

where ω denotes the segmentation rate. Moreover, let the size of X_{3D} be $x_1 \times y_1 \times \lambda_1$, thus the sizes of X_{cen} and X_{nbr} are given as $x_2 \times \lambda_1$ and $x_3 \times \lambda_1$, respectively, where $x_2 + x_3 = x_1 \times y_1$.

The obtained X_{cen} and X_{nbr} are then used as inputs to the VBSST encoder. As shown in Fig. 4, the VBSST encoder mainly consists of a multi-head attention module and a feed-forward network (FFN). Regarding the multi-head attention module, it takes three inputs: query (denoted as Q), key (denoted as K), and value (denoted as V), which are used to compute the single-head attention as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

$$\text{head}_i = \text{Attention}(Q, K, V) \quad (3)$$

where head_i denotes the single-head attention, d_k denotes the scale factor, and its size is determined by the dimensionality of the query. As can

be seen from (2), the attention mechanism is implemented by calculating the dot product between the query and the key. The result of the dot product is then used as weights for the value through the Softmax function and the scale factor.

Moreover, the multi-head attention is defined as the concatenation of single-head attentions, as follows:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) w^o \quad (4)$$

where w^o represents the weight matrix of the linear transformation.

Inspired by the concept of local binary pattern [50], we select X_{cen} as the query, while X_{nbr} is used as both the key and the value. This approach fully exploits the neighboring correlations by computing the difference distribution within the local neighborhood, as follows:

$$Q'_{img} = w_i^q X_{cen} \quad (5)$$

$$K'_{img} = w_i^k X_{nbr} \quad (6)$$

$$V'_{img} = w_i^v X_{nbr} \quad (7)$$

where w_i^q , w_i^k , and w_i^v denote the weight matrices for the query, key, and value in the i th group, respectively.

Hence, the multi-head attention mechanism, incorporating neighboring correlations for visual features, is defined as follows:

$$\text{MultiHead}_{img} = \text{MultiHead}(Q'_{img}, K'_{img}, V'_{img}) \quad (8)$$

Then, the multi-attention scores are passed through the FFN, together with the central sequence X_{cen} , for computing the visual spatial-spectral (VSS) feature tensor F_{img} . The FFN consists of two fully connected layers, with ReLU as the activation function in the first layer and a linear transformation in the second layer. The VSS feature tensor F_{img} is expressed as follows:

$$\begin{aligned} F_{img} &= \text{FFN}(\text{Ln}(X_{cen} + \text{MultiHead}_{img})) \\ &= \max\{0, ((\text{Ln}(X_{cen} + \text{MultiHead}_{img})) w_1 + b_1) w_2 + b_2\} \end{aligned} \quad (9)$$

where w_1 , b_1 and w_2 , b_2 are the weights and biases of the first and second fully connected layers, respectively, and $\text{Ln}(\cdot)$ denotes layer normalization. Finally, the VSS feature tensor are flattened into a one-dimensional vector, and the embedding function for the VBSST encoder is formulated as follows:

$$f_{img}(X_{3D}) = \text{flatten}(F_{img}) \quad (10)$$

In the case of the TBST encoder, some components are similar to those in the VBSST encoder, but there are still some differences between them, which will be explained below. First, the text X_{test} is tokenized and then transformed into vectors using methods such as byte-pair encoding (BPE), as defined below:

$$X_{emb} = \text{Encoding}(X_{test}) \quad (11)$$

where $\text{Encoding}(\cdot)$ denotes the encoding function of BPE, X_{emb} is the textual vector after encoding, and positional embeddings are then added to it, as follows:

$$X'_{emb} = X_{emb} + \text{position}(X_{emb}) \quad (12)$$

where X'_{emb} denotes the textual vector after positional encoding. For the self-attention module of the TBST encoder, Q'_{text} , K'_{text} and V'_{text} are derived from the linear mapping of X'_{emb} and are computed in a manner similar to that in Eqs. (5), (6), and (7). Additionally, the multi-head attention mechanism for textual features MultiHead_{text} follows the same calculation as described in Eqs. (4) and (8).

Unlike the VBSST encoder, the TBST encoder consists of multiple residual blocks (ResBlocks). A ResBlock is computed as follows:

$$\text{resblock}(X_{text}) = \text{Ln}(\text{FFN}(\text{Ln}(Q'_{text} + \text{MultiHead}_{text}))) \quad (13)$$

Then, the textual semantic features are obtained through L stacked ResBlocks, and the embedding function of the TBST encoder is defined as follows:

$$f_{text}(X_{text}) = \text{Stack}(\text{resblock}_1, \text{resblock}_2, \dots, \text{resblock}_L) \quad (14)$$

After obtaining the visual-textual spatial-spectral features, we introduce a visual-linguistic-prototype alignment method to optimize the parameters of VTDFormer. As shown in Figs. 2 and 4, this method consists of two stages: visual-prototype alignment and visual-linguistic alignment. In the visual-prototype alignment, each episode is divided into a support set and a query set:

$$S_s = \{(x_i, y_i)\}_{i=1}^{C \times K} \quad (15)$$

$$Q_s = \{(x_j, y_j)\}_{j=1}^{C \times N} \quad (16)$$

where S_s denotes the support set consisting of C classes, with K samples per class, and Q_s denotes the query set consisting of C classes, with N samples per class.

Particularly, we compute the prototype from the support set as follows:

$$c_k = \frac{1}{|S_s|} \sum_{(x_i, y_i) \in S_s} f_{img}^\lambda(x_i) \quad (17)$$

where c_k denotes the prototype of k th class. Then, the visual-prototype alignment is performed by computing the distance between the VSS features of query samples and the prototypes. Therefore, the class probability for a query sample $x_j \in Q_s$ during the visual-prototype alignment stage can be defined as follows:

$$p_{sup-que}(y_j = k | x_j \in Q_s) = \frac{\exp(-d(f_{img}^\lambda(x_j), c_k))}{\sum_{k=1}^C \exp(-d(f_{img}^\lambda(x_j), c_k))} \quad (18)$$

where $d(\cdot)$ denotes the Euclidean distance function, $\exp(\cdot)$ denotes the exponential function, f_{img}^λ denotes the embedding function for the VBSST encoder with parameter λ , and y_j is the label of x_j .

Based on the negative log-probability of the query sample x and its corresponding class label k , the visual-prototype alignment loss for all query samples in a source or target episode can be expressed as:

$$L_{v-p}^s = E_{S_s, Q_s} \left[- \sum_{(x, y) \in Q_s} \log p_{sup-que}(y = k | x) \right] \quad (19)$$

$$L_{v-p}^t = E_{S_t, Q_t} \left[- \sum_{(x, y) \in Q_t} \log p_{sup-que}(y = k | x) \right] \quad (20)$$

where S_t and Q_t are the support and query sets from the target-domain dataset, respectively.

In the visual-linguistic alignment stage, each episode is represented as the combination of the support set and the query set.

$$\text{Img}_s = S_s \cup Q_s = \{(x_u, y_u)\}_{u=1}^{C \times (K+N)} \quad (21)$$

where Img_s denotes the episode consisting of C classes, with $K + N$ samples per class in the visual-linguistic alignment stage.

To utilize the domain invariance of the shared linguistic domain, a distance metric is further computed between the visual spatial-spectral features and the concept-guided textual features. The class probability for an HSI sample $x_j \in \text{Img}_s$ during this phase can be defined as follows:

$$p_{img-text}(y_j = k | x_j \in \text{Img}_s) = \frac{\exp(-d(f_{img}^\lambda(x_j), f_{text}^\beta(\text{text}_k)))}{\sum_{k=1}^C \exp(-d(f_{img}^\lambda(x_j), f_{text}^\beta(\text{text}_k)))} \quad (22)$$

where $text_k$ denotes the C2TD corresponding to c_k , and f_{text}^β denotes the embedding function for the TBST encoder with parameter β .

Similarly, based on the negative log-probability of the HSI sample x and its corresponding class label k , the visual-linguistic alignment loss for all HSI samples in a source or target episode can be calculated as follows:

$$L_{v-t}^s = E_{Img_s, text_s} \left[- \sum_{(x,y) \in Img_s} \log p_{img-text}(y = k|x) \right] \quad (23)$$

$$L_{v-t}^t = E_{Img_t, text_t} \left[- \sum_{(x,y) \in Img_t} \log p_{img-text}(y = k|x) \right] \quad (24)$$

Finally, the visual-linguistic-prototype alignment loss for the VTDFormer is computed as follows:

$$L_{align} = L_{v-p}^s + L_{v-p}^t + L_{v-t}^s + L_{v-t}^t \quad (25)$$

3.3. Spatial-spectral-textual domain adversarial module (SSTDAM)

The proposed SSTDAM introduces two complementary domains: the spatial-spectral (visual) domain and the linguistic domain. The spatial-spectral domain further comprises both source and target subdomains, which are jointly involved in the adversarial training process. As depicted in Figs. 1 and 5, SSTDAM adopts a dual-domain adversarial learning framework that simultaneously operates across the source-target and visual-linguistic spaces. Specifically, the visual-linguistic adversarial branch facilitates the alignment between concept-embedded textual representations and spatial-spectral visual features, while the source-target adversarial branch enforces the learning of domain-invariant visual representations. By jointly optimizing these two adversarial objectives, SSTDAM effectively reduces cross-domain semantic discrepancies and enhances the overall generalization capability of the proposed CGVT-FSL model.

First, a source-target discriminator D_{s-t} is introduced to address the domain shift between the source distribution $P_s(x)$ and the target distribution $P_t(x)$. The classifier prediction g_{img} for the VSS features captures discriminative information, which is refined through adversarial adaptation of the corresponding feature representation f_{img} . During adversarial learning, the domain variances of both f_{img} and g_{img} are simultaneously modeled. Consequently, the source-target domain adversarial loss function L_{s-t} is defined on the domain discriminator D_{s-t} , as follows:

$$\begin{aligned} \min_{D_{s-t}} \max_{f_{img}, g_{img}} L_{s-t} &= -E_{x_i^s \sim P_s(x)} \log [D_{s-t}(f_{img}^s, g_{img}^s)] \\ &\quad - E_{x_j^t \sim P_t(x)} \log [1 - D_{s-t}(f_{img}^t, g_{img}^t)] \end{aligned} \quad (26)$$

where $D_{s-t}(\cdot)$ computes the probability that x_i^s belongs to the source domain, and $1 - D_{s-t}(\cdot)$ computes the probability that x_j^t belongs to the target domain. The source-target domain discriminator $D_{s-t}(\cdot)$ minimizes the loss function, while the VSS feature representation f_{img} and the classifier prediction g_{img} for the corresponding features maximize the adversarial loss.

Similarly, the adversarial loss for the visual-linguistic domain discriminator $D_{img-text}$ is computed as follows:

$$\begin{aligned} \min_{D_{img-text}} \max_{f_{img}, g_{img}, f_{text}, g_{text}} L_{img-text} &= -E_{Img_i \sim P_{img}(x)} \log [D_{img-text}(f_{img}, g_{img})] \\ &\quad - E_{text_j \sim P_{text}(x)} \log [1 - D_{img-text}(f_{text}, g_{text})] \end{aligned} \quad (27)$$

where $D_{img-text}(\cdot)$ computes the probability that Img_i belongs to the visual domain, and $1 - D_{img-text}(\cdot)$ computes the probability that $text_j$ belongs to the linguistic domain. $P_{img}(x)$ and $P_{text}(x)$ represent the visual and linguistic distributions, respectively. Hence, the visual-linguistic domain discriminator $D_{img-text}(\cdot)$ minimizes the loss $L_{img-text}$, while the visual and textual spatial-spectral feature representations f_{img} and f_{text} , and the classifier predictions g_{img} and g_{text} for the corresponding features maximize the adversarial loss.

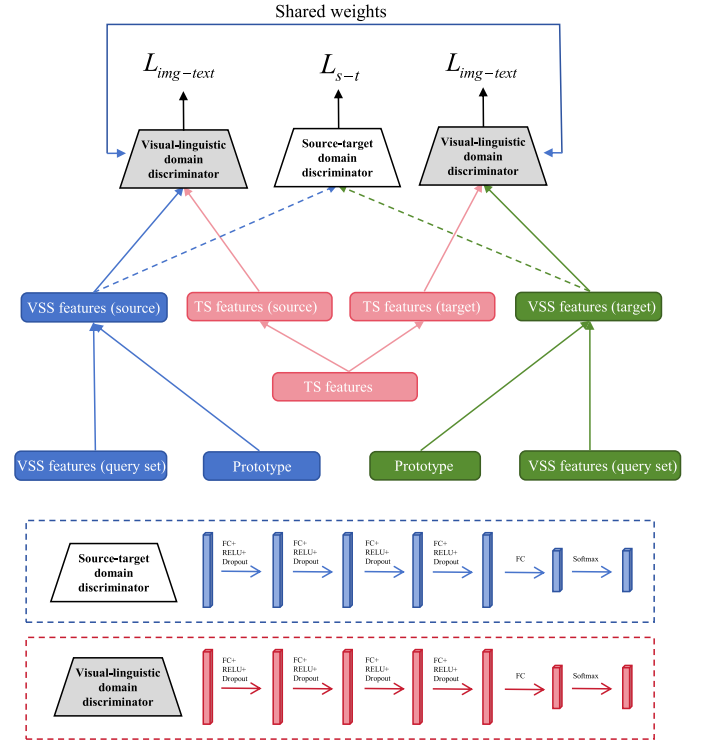


Fig. 5. Illustration framework of the proposed SSTDAM.

Let $h = (f, g)$ represent the joint variable of f and g . As described in [15], multilinear mapping is performed between f and g for domain adversarial learning, which is defined as the outer product of multiple random vectors, i.e., $f \otimes g$. This approach may lead to dimensionality explosion. Therefore, we adopt the following conditioning strategy to mitigate this issue:

$$T(h) = \begin{cases} f \otimes g, d_f \times d_g \leq 1024 \\ (1/\sqrt{d})(R_f f) \odot (R_g g), \text{otherwise} \end{cases} \quad (28)$$

where d_f and d_g represent the dimensions of f and g , $\odot$ denotes the element-wise product, $R_f \in R^{d \times d_f}$ and $R_g \in R^{d \times d_g}$ are two random matrices that are sampled only once and fixed during the training phase, with $d \ll d_f \times d_g$. If the dimension of the multilinear mapping exceeds the maximum number of units in the deep network (i.e., 1024), the multilinear mapping can be approximated by the dot-product $(1/\sqrt{d})(R_f f) \odot (R_g g)$.

Therefore, the adversarial loss for the source-target domain discriminator is transformed into L'_{s-t} , as follows:

$$\begin{aligned} L'_{s-t} &= \min_{D_{s-t}} \max_T L_{s-t} = -E_{x_i^s \sim P_s(x)} \log [D_{s-t}(T(h_i^s))] \\ &\quad - E_{x_j^t \sim P_t(x)} \log [1 - D_{s-t}(T(h_j^t))] \end{aligned} \quad (29)$$

Similarly, the adversarial loss for the visual-linguistic domain discriminator is reformulated as follows:

$$\begin{aligned} L'_{img-text} &= \min_{D_{img-text}} \max_T L_{img-text} = -E_{Img_i \sim P_{img}(x)} \log [D_{img-text}(T(h_i^{img}))] \\ &\quad - E_{text_j \sim P_{text}(x)} \log [1 - D_{img-text}(T(h_j^{text}))] \end{aligned} \quad (30)$$

Moreover, we use a multilayer perceptron (MLP) placed after a multilinear mapping. The MLP consists of five fully connected layers, with a ReLU activation function and dropout layers applied after each fully connected layer, except for the final output layer. The output layer employs a Softmax activation function, which is used to classify whether

the input data originates from the source domain (visual domain) or the target domain (linguistic domain).

Finally, the adversarial loss from the proposed SSTDAM is denoted as the weighted sum of L'_{s-t} and $L'_{img-text}$. In the proposed CGVT-FSL, the VTDFormer and the SSTDAM are trained jointly. Therefore, the total loss of the CGVT-FSL method is computed as follows:

$$L_{total} = L_{align} + L'_{s-t} + \lambda L'_{img-text} \quad (31)$$

where λ denotes the weighting coefficient of the visual-linguistic domain adversarial loss.

4. Experiment

Experiments are conducted on a computer with an Intel Xeon Silver 4130 processor (2.1 GHz), 32 GB of DDR4 RAM, and an NVIDIA GeForce RTX 4090 graphics processing unit (GPU). The training and testing phases are implemented using the open-source PyTorch framework.

In the experimental setup, we employed three separate Adam optimizers to optimize the VTDFormer, the Source-target domain discriminator, and the Visual-linguistic domain discriminator, respectively, with a uniform learning rate of 0.001. In each training episode, we alternated between using source domain and target domain data for training. In every iteration, the corresponding losses were computed and back-propagated, while the parameters of the three networks were updated simultaneously. The entire training process consisted of 10,000 episodes, with training loss and accuracy output every 100 episodes, and model testing conducted every 1000 episodes. No learning rate decay strategy was employed during training, and the weight decay parameter of the optimizers was kept at the default value of 0.

4.1. Experimental dataset

To evaluate the performance of CGVT-FSL, six public HSI datasets are used, including Chikusei [51], Houston (HT) [52], Indian Pines (IP) [53], Pavia University (PU) [53], WHU-Hi-HanChuan (HC) [54] and WHU-Hi-HongHu (HH) [54] datasets. Since CGVT-FSL is a cross-domain few-shot learning approach, the Chikusei dataset is selected as the source domain dataset, while the HT, the IP, the PU, the HC, and the HH datasets are used as target domain datasets. Notably, the HC dataset contains a large number of samples, so 15% of the samples in each class are randomly selected. Similarly, since the HH dataset also contains a large number of samples, only the first 18 classes are considered, with 10% of the samples from each class randomly selected. Detailed information about the experimental datasets is provided in Table 1, and Fig. 6 shows the false-color images and corresponding ground-truth maps of the five target domain datasets.

4.2. Experimental setting

To verify the effectiveness of our proposed method, we compare the CGVT-FSL method with three supervised methods (i.e., SVM [55], 3D-CNN [37], and SSRN [38]), and seven FSL methods (i.e., DFSL [14], DCFSL [15], CMFSL [56], Gia-CFSL [42], MRLF [43], HyMuT [44] and SGPL [8]). In terms of FSL methods, a cross-domain episodic training strategy is employed during the training phase. Specifically, 200 labeled

samples per class from the source domain are selected to learn transferable knowledge, while a limited number of labeled samples from the target domain are used for training. The remaining samples from the target domain datasets are reserved for testing.

In the experiments, the spatial size of the 3-D local tensor is set to 9×9 , while the spectral dimension is fixed at 128. All network parameters are initialized using the Kaiming initialization scheme. Moreover, the TBST encoder is initially configured as a three-ResBlock transformer model with approximately 33 M parameters and eight attention heads. It adopts a CLIP-like text encoder architecture, where textual inputs are tokenized using lowercase byte pair encoding (BPE) with a vocabulary size of 49,152. To ensure computational efficiency, the maximum sequence length is limited to 76.

To comprehensively evaluate the classification performance, three widely adopted metrics in hyperspectral remote sensing are employed, including Overall Accuracy (OA), Average Accuracy (AA), and Kappa coefficient (Kappa) [57]. Specifically, OA measures the proportion of correctly classified pixels among all test samples, providing a global micro-averaged assessment. AA is calculated as the mean accuracy across all classes, offering a macro-averaged and class-balanced evaluation, which is particularly meaningful for imbalanced datasets. Kappa quantifies the agreement between the predicted results and ground truth while accounting for random chance. A larger Kappa value indicates stronger classification reliability beyond random agreement. Together, these metrics provide a comprehensive and complementary evaluation of the model performance.

4.3. Parameters tuning

A parameter sensitivity analysis is conducted to assess the sensitivity of CGVT-FSL to changes in the weighting coefficient λ for the visual-linguistic domain discriminator. The values of the weighting coefficient λ are set to 0.9, 0.95, 1.0, 1.05, and 1.1 for the analysis.

After adjustment by the weighting coefficient λ , the loss gradient is used in stochastic gradient descent to update model parameters. Table 2 presents the classification results of CGVT-FSL with different values of the weighting coefficient λ across five target domain datasets. It can be observed that the optimal performance of CGVT-FSL for the HT dataset is achieved when λ is 1.10, suggesting that a higher weight should be assigned to the visual-linguistic domain adversarial loss. However, for the IP, PU, HC and HH datasets, the optimal performance of CGVT-FSL occurs when λ is set to 1.0, corresponding to a lower weight for the visual-linguistic domain adversarial loss.

This discrepancy arises because the HT datasets contain many category pairs with minimal semantic differences (e.g., Parking_lot and Parking_lot2), resulting in highly similar class-label text descriptions for certain categories. As a result, the model needs to explore and leverage textual information more thoroughly, which accounts for the need for a larger value of λ . In contrast, other datasets contain fewer such pairs, and when λ is set to 1.0, the text information has already been adequately leveraged to enhance the generalization performance of the model. In summary, setting λ to 1.0 allows the proposed CGVT-FSL to achieve optimal performance across the five target domain datasets. Therefore, in the following experiment, we fix λ at 1.0.

Table 1
All details of the experimental datasets.

Dataset	Location	Resolution	Spectral Band Range	Band Number	Class Number	Label Samples	Sensor Type
Chikusei	Japan	2517 × 2335	363–1018 nm	128	19	77,592	Hyperspec-VNIR-C
HT	USA	349 × 1905	380–1050 nm	144	15	15,029	ITRES CASI-1500
IP	USA	145 × 145	400–2500 nm	200	16	10,249	AVIRIS
PU	Italy	610 × 340	430–860 nm	103	9	42,776	ROSIS
HC	China	1217 × 303	400–1000 nm	274	16	38,622	Headwall Nano-Hyperspec
HH	China	940 × 475	400–1000 nm	270	18	36,905	Headwall Nano-Hyperspec

Note: “Label Samples” refers to the number of labeled pixels used for training, excluding the background class.

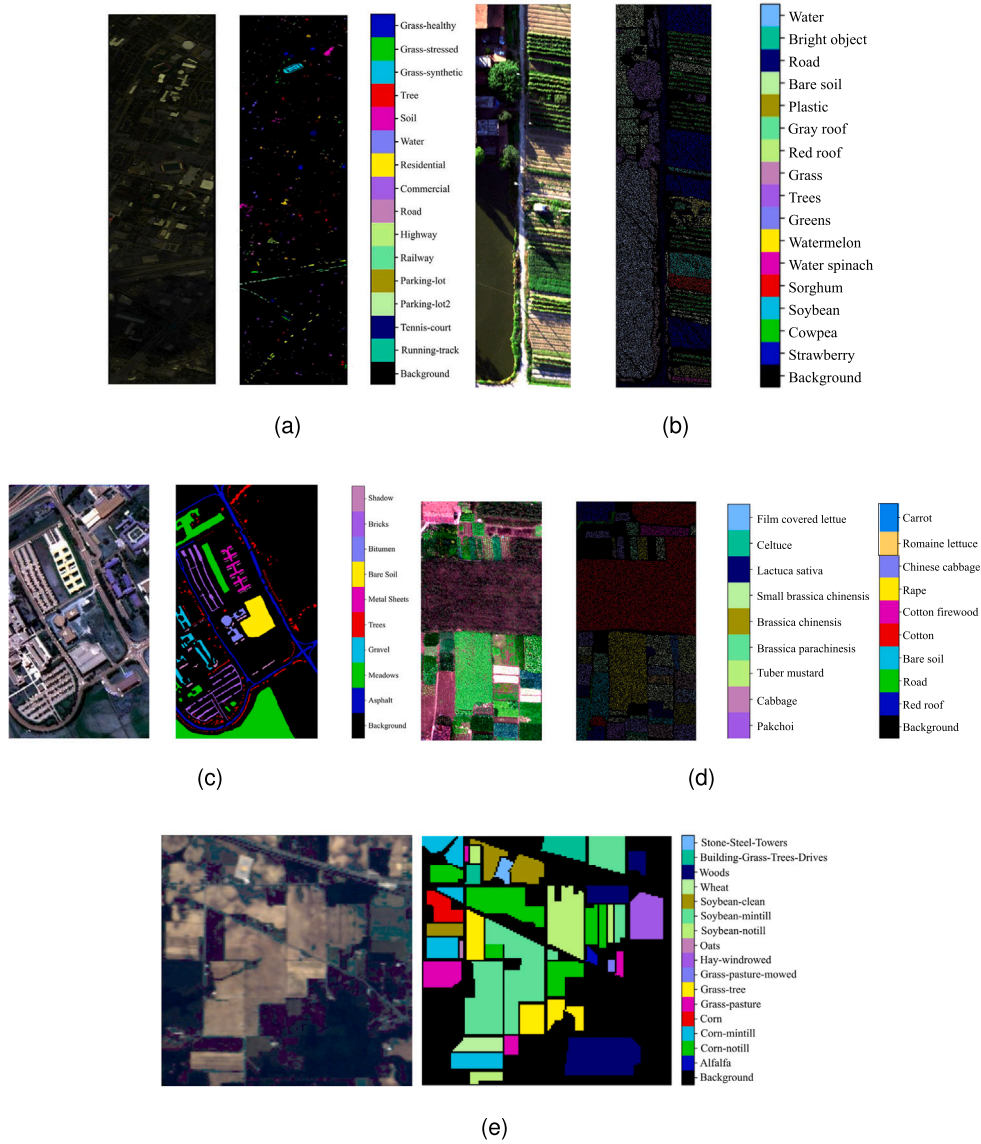


Fig. 6. False-color images and ground-truth maps of the five target domain datasets: (a) HT, (b) HC, (c) PU, (d) HH and (e) IP datasets.

Table 2

Overall Accuracy (%) of CGVT-FSL with different values of the weighting coefficient λ in Five-Shot classification for Five target domain Datasets.

λ	HT	IP	PU	HC	HH	Mean
0.90	83.02 $\pm$ 2.67	76.16 $\pm$ 2.54	87.30 $\pm$ 5.31	77.23 $\pm$ 3.21	85.64 $\pm$ 2.33	81.87 $\pm$ 3.21
0.95	83.44 $\pm$ 2.67	76.30 $\pm$ 2.54	87.70 $\pm$ 5.31	77.21 $\pm$ 3.22	85.90 $\pm$ 1.93	82.11 $\pm$ 3.13
1.00	83.35 $\pm$ 2.49	77.68 $\pm$ 2.14	88.34 $\pm$ 2.09	77.29 $\pm$ 3.00	86.06 $\pm$ 2.16	82.54 $\pm$ 2.38
1.05	83.20 $\pm$ 2.52	76.60 $\pm$ 2.21	87.98 $\pm$ 5.78	77.12 $\pm$ 3.42	86.02 $\pm$ 1.69	82.18 $\pm$ 3.12
1.10	83.45 $\pm$ 2.52	77.57 $\pm$ 3.04	88.32 $\pm$ 5.06	77.10 $\pm$ 3.22	85.99 $\pm$ 1.96	82.49 $\pm$ 3.16

4.4. Ablation study

In this section, as shown in Table 3, an ablation study is conducted to assess the contributions of each component of the proposed CGVT-FSL. Specifically, we systematically remove key components corresponding to three critical phases in cross-domain HSI classification: feature pre-processing, feature extraction, and domain adversarial learning. Based on these removals, three degraded variants of CGVT-FSL are constructed as follows:

1. “CGVT-FSL (No C2TD)”: In the feature preprocessing phase, this variant excludes the C2TD, disabling the integration of concept-level semantics and the modeling of intra-class and inter-class relationships in class-label text.
2. “CGVT-FSL (No VTDFormer)”: In the feature extraction phase, this variant removes the TBST encoder, disabling cross-domain spatial-spectral feature extraction in the textual branch.

Table 3

Overall Accuracy (%) of CGVT-FSL (No C2TD), CGVT-FSL (No VTDFormer), CGVT-FSL (No SSTDAM), and CGVT-FSL on Five target domain datasets using Five labeled samples per Class.

Method	CGVT-FSL (No C2TD)	CGVT-FSL (No VTDFormer)	CGVT-FSL (No SSTDAM)	CGVT-FSL
HT	83.24 ± 2.63	83.14 ± 2.10	83.30 ± 2.66	83.35 ± 2.49
IP	77.12 ± 2.79	70.66 ± 3.29	77.31 ± 2.82	77.68 ± 2.14
PU	87.40 ± 5.37	84.67 ± 4.20	86.71 ± 4.99	88.34 ± 2.09
HC	76.59 ± 3.26	76.76 ± 3.35	76.99 ± 3.66	77.29 ± 3.00
HH	85.55 ± 2.18	85.23 ± 2.68	85.80 ± 2.27	86.06 ± 2.16
Mean	81.98 ± 3.25	80.09 ± 3.12	82.02 ± 3.28	82.54 ± 2.38

Table 4

Overall Accuracy(%) of CGVT-FSL(CNO), CGVT-FSL(CNC), CGVT-FSL(STO) and CGVT-FSL(STC) on PU and IP datasets using Five labeled samples per class.

Style	Standard Template	Concepts	PU	IP
CGVT-FSL(CNO)	–	–	86.28 ± 5.62	76.96 ± 2.43
CGVT-FSL(CNC)	–	✓	87.90 ± 5.47	77.06 ± 3.28
CGVT-FSL(STO)	✓	–	87.40 ± 5.37	77.12 ± 2.79
CGVT-FSL(STC)	✓	✓	88.34 ± 2.09	77.68 ± 2.14

3. “CGVT-FSL (No SSTDAM)”: In the domain adversarial learning phase, this variant removes the visual-linguistic domain discriminator, disabling adversarial alignment between the visual and linguistic domains.

Our proposed C2TD module relies on prompt-based concept generation to enhance cross-modal understanding of HSIs. To thoroughly evaluate the effectiveness of the C2TD module, we systematically investigate the impact of different prompting styles on the performance of the CGVT-FSL model, considering four combinations: CNO (Class Names Only), CNC (Class Names with Concepts), STO (Standard Template Only), and STC (Standard Template with Concepts). The corresponding experimental results are presented in Table 4.

As shown in Table 4, the CGVT-FSL(STC) consistently outperforms all other styles, confirming the effectiveness of the standard template and concepts. First, CGVT-FSL(CNO) vs. CGVT-FSL(CNC) and CGVT-FSL(STO) vs. CGVT-FSL(STC), we observe that incorporating descriptive concepts (e.g., “[Agricultural]”) into the prompts consistently improves model performance. This validates that our proposed concept-guided prompting provides richer semantic context, facilitating more precise cross-modal alignment. Second, comparisons between CGVT-FSL(CNO) vs. CGVT-FSL(STO) and CGVT-FSL(CNC) vs. CGVT-FSL(STC) indicate that including a standard template like “A hyperspectral image of Class-name” yields additional performance gains. This suggests that a well-structured natural language template helps the TBST better understand the input and generate more discriminative feature representations. Therefore, in the following experiment, we chose STC as the prompting style of CGVT-FSL.

Building upon the STC prompting style identified above, we further evaluate the contribution of VTDFormer to the shared feature representation. As summarized in Table 5, the full model with VTDFormer demonstrates consistent advantages across the IP, HH, and PU datasets. Specifically, it achieves higher Prototype Accuracy (Proto Acc), smaller Positive Prototype Distance (Pos Proto Dist), and larger Prototype Margin (Proto Margin) on all three datasets. These results indicate that VTDFormer promotes tighter intra-class clustering while improving inter-class separability in the feature space. This enhanced structure leads to a more discriminative and well-organized prototype representation. Therefore, the results in Table 5 confirm that the role of VTDFormer is not merely to increase model complexity, but to improve the structural quality of the shared representation space for cross-domain few-shot classification.

Table 5

Merged summary of the favorable representation-level results of VTDFormer across three datasets.

Dataset	Metric	Model (no VTDFormer)	Full Model (with VTDFormer)
IP	Proto Acc	69.32% ± 4.07%	86.93% ± 2.51%
	Pos Proto Dist	0.5841 ± 0.0314	0.5747 ± 0.0461
	Proto Margin	0.2776 ± 0.0401	1.0737 ± 0.0870
HH	Proto Acc	68.03% ± 6.38%	69.42% ± 5.89%
	Pos Proto Dist	0.4392 ± 0.0615	0.4135 ± 0.0628
	Proto Margin	0.0963 ± 0.0906	0.1201 ± 0.0840
PU	Proto Acc	47.24% ± 4.32%	48.73% ± 3.54%
	Pos Proto Dist	0.6486 ± 0.0768	0.6280 ± 0.0749
	Proto Margin	−0.2937 ± 0.0774	−0.2696 ± 0.0744

To further assess the contribution of SSTDAM, we analyze both feature-space visualization and discrepancy metrics. As shown in Fig. 7, the model with SSTDAM produces a more compact and well-structured feature space compared with the variant without SSTDAM. This observation is consistent with the quantitative results in Table 6, where both Maximum Mean Discrepancy (MMD) [58] and Correlation Alignment (CORAL) [59] are significantly reduced after introducing SSTDAM. These results demonstrate that SSTDAM effectively mitigates source–target discrepancy and improves cross-domain alignment.

4.5. Compared with other training paradigms

This experiment investigates different training paradigms for FSL, comparing three distinct approaches:

1. SGPL: The model is first pre-trained on the source domain. Then, the entire network (not just the final layer) is fine-tuned using a limited number of labeled samples from the target domain.
2. Linear Probing (Modified CGVT-FSL): This approach is a simplified variant derived from the full CGVT-FSL framework. It involves pre-training the model on the source domain, followed by fine-tuning only a linear classification layer with the few labeled target samples.
3. CGVT-FSL: Our complete framework employs an alternating training strategy between the source and target domains. It integrates adversarial domain alignment, specifically through the SSTDAM module, to explicitly reduce domain shift and achieve joint adaptation of the feature spaces.

As shown in Table 7, under the 5-shot setting, the proposed CGVT-FSL framework consistently outperforms both the Linear Probe baseline and the SGPL method on the HC and HH datasets. Specifically, compared with Linear Probe, CGVT-FSL improves OA by 4.85% and 2.84%, AA by 6.77% and 5.81%, and the Kappa coefficient by 5.48% and 4.49% on HC and HH, respectively. These results indicate that simply fine-tuning a linear classifier on fixed features is insufficient for cross-domain adaptation, while the proposed framework effectively enhances feature transferability.

The computational cost comparison in Table 8 shows that CGVT-FSL requires longer training time than the Linear Probe baseline on both datasets. However, the two methods exhibit nearly identical inference costs, with the same FLOPs (3.019 G) and very similar inference times. This is because the dominant computational cost is determined by the shared visual–textual backbone, while the additional components in CGVT-FSL contribute only marginally to the inference complexity.

In contrast, the difference in trainable parameters (39.42 M vs 43.61 M) reflects the additional modules introduced for cross-domain semantic alignment in CGVT-FSL. These components mainly affect the training stage rather than the inference-time computation. Compared

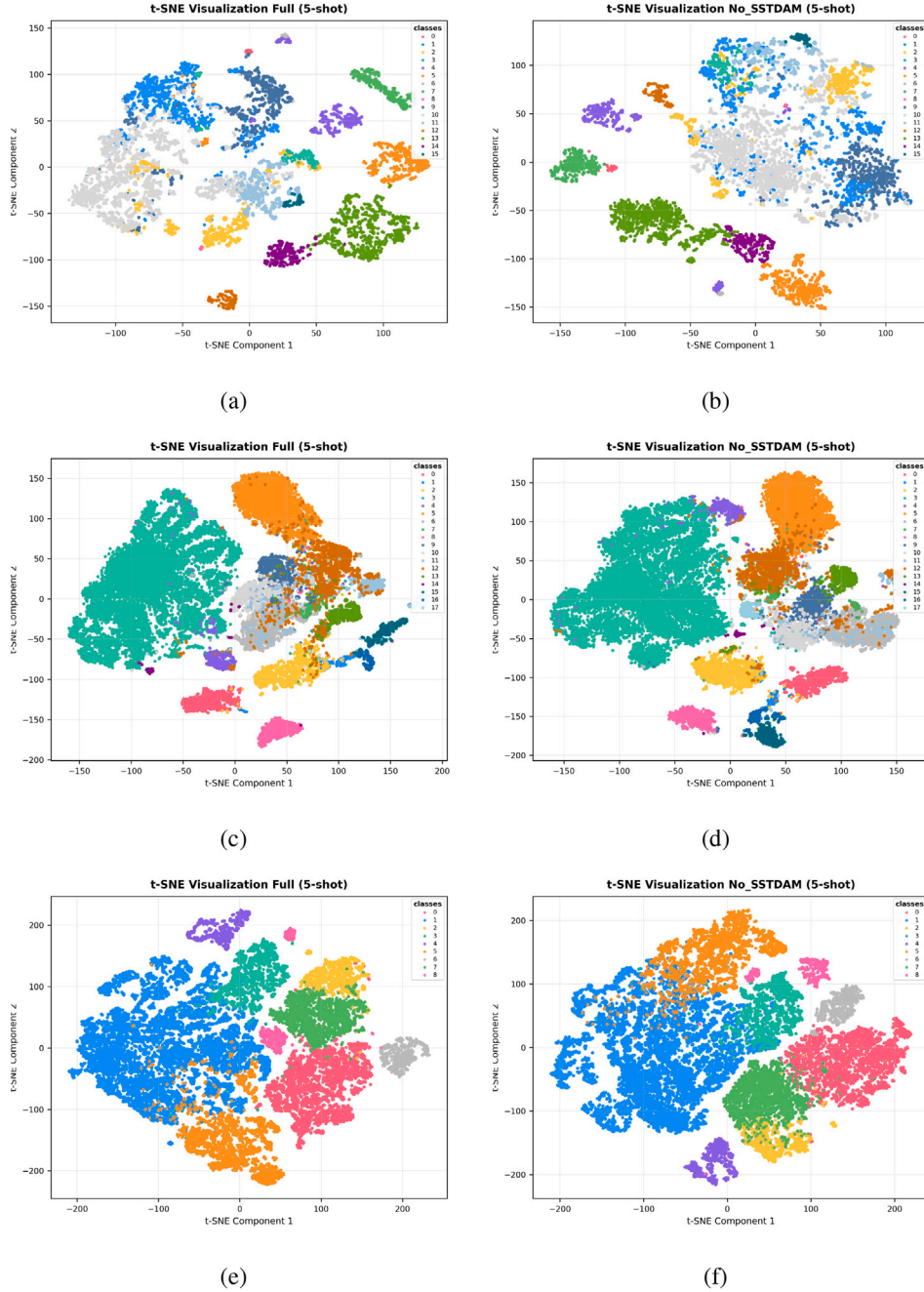


Fig. 7. t-SNE visualizations of the shared feature space on the IP, HH, and PU datasets for the full model and the model without SSTDA. Subfigures correspond to: (a) IP, full model; (b) IP, no SSTDA; (c) HH, full model; (d) HH, no SSTDA; (e) PU, full model; and (f) PU, no SSTDA.

Table 6

Quantitative comparison of source–target discrepancy with and without SSTDA.

Dataset	Metric	Model (no SSTDA)	Full Model (with SSTDA)
IP	CORAL	1.11 ± 0.093	0.258 ± 0.067
	MMD	0.145 ± 0.017	0.050 ± 0.014
HH	CORAL	7.59 ± 0.679	3.33 ± 0.976
	MMD	0.175 ± 0.030	0.090 ± 0.026
PU	CORAL	0.884 ± 0.090	0.250 ± 0.056
	MMD	0.113 ± 0.017	0.045 ± 0.015

with SGPL, CGVT-FSL maintains comparable deployment efficiency while achieving superior classification performance.

Overall, these results demonstrate that CGVT-FSL achieves improved cross-domain performance without incurring additional inference-time cost, indicating a favorable trade-off between effectiveness and efficiency.

4.6. Prompt-construction ablation

To further clarify the distinction between C2TD and ordinary prompt enrichment, we conduct a prompt-construction ablation study

Table 7

Comparison of OA, AA, and Kappa (%) for SGPL, Linear Probe, and CGVT-FSL on the HC and HH Datasets.

Method	Metric	HC	HH
SGPL	OA	67.35 $\pm$ 0.61	82.88 $\pm$ 1.17
	AA	55.99 $\pm$ 1.10	67.19 $\pm$ 0.73
	Kappa	62.18 $\pm$ 0.61	78.27 $\pm$ 1.36
Linear Probe	OA	72.44 $\pm$ 4.26	83.22 $\pm$ 4.09
	AA	68.16 $\pm$ 2.59	76.10 $\pm$ 2.71
	Kappa	68.33 $\pm$ 4.63	77.76 $\pm$ 4.81
CGVT-FSL	OA	77.29 $\pm$ 3.00	86.06 $\pm$ 2.16
	AA	74.93 $\pm$ 2.48	81.91 $\pm$ 2.30
	Kappa	73.81 $\pm$ 3.30	82.25 $\pm$ 2.56

Table 8

Computational cost comparison on HC and HH.

Method	Metric	HC	HH
SGPL	Training Time	1716.84 s	1562.47 s
	Inference Time	9.45 s	9.23 s
	Trainable Params	37.31 M	37.33 M
	FLOPs	2.955 G	2.955 G
Linear Probe (Baseline)	Training Time	1284.36 s	1173.95 s
	Inference Time	8.12 s	8.04 s
	Trainable Params	39.42 M	39.43 M
	FLOPs	3.019 G	3.019 G
CGVT-FSL	Training Time	2053.28 s	1846.53 s
	Inference Time	8.09 s	7.99 s
	Trainable Params	43.61 M	43.63 M
	FLOPs	3.019 G	3.019 G

on two representative target datasets, PU and IP. Five prompt-construction strategies are compared, including Class-Name Only, C2TD with Shuffled Concepts, Class-Specific Rich Description, Scene-Context Relation Description, and C2TD with Correct Concepts.

Specifically, four controlled prompt baselines are introduced to distinguish the effect of semantically correct concept-level guidance from that of simply using longer, richer, or more context-enriched textual descriptions. Class-Name Only removes all concept labels and additional descriptive information, and retains only the class name as the category-specific semantic content, e.g., “Alfalfa”. In implementation, the class name is wrapped by the same basic HSI template for all methods to ensure a consistent input format. Class-Specific Rich Description uses manually written category-specific descriptions for each class, but does not include the structured C2TD concept labels, e.g., “Alfalfa, a small agricultural forage crop class with vegetation reflectance and sparse field samples”. Scene-Context Relation Description further introduces scene-dependent inter-class contextual information, such as adjacency, transition, and spatial co-occurrence relationships, e.g., “Alfalfa, a sparse forage crop class that may appear near other agricultural parcels such as Corn-notill, Soybean-notill, and Grass-pasture in the scene”. These two settings serve as stronger prompt-enrichment baselines beyond simple class-name prompts.

In addition, C2TD with Shuffled Concepts preserves the same textual format and concept vocabulary as C2TD with Correct Concepts, but randomly mismatches the concept labels with the class names. For example, the correct C2TD semantic content for Alfalfa is “Alfalfa [Agricultural] [Medium density] [Low specific heat capacity]”, while the shuffled counterpart may be “Alfalfa [Natural] [Low density] [High specific heat capacity]”. This setting controls the prompt length, structure, and concept vocabulary, while destroying the semantic correctness of class-to-concept matching. Therefore, these variants are used to examine whether the improvement of C2TD comes from semantically correct concept-level guidance rather than prompt length, prompt format, richer

Table 9

Prompt-construction ablation results on the PU and IP datasets.

Method	Metric	PU	IP
Class-Name Only	OA	87.40 $\pm$ 5.37	76.96 $\pm$ 2.43
	AA	88.48 $\pm$ 1.65	85.61 $\pm$ 1.89
	Kappa	81.79 $\pm$ 4.92	74.14 $\pm$ 3.12
C2TD with Shuffled Concepts	OA	86.46 $\pm$ 5.18	76.68 $\pm$ 3.00
	AA	89.92 $\pm$ 1.71	85.87 $\pm$ 1.49
	Kappa	82.70 $\pm$ 5.99	73.68 $\pm$ 3.29
Class-Specific Rich Description	OA	87.25 $\pm$ 5.09	77.01 $\pm$ 2.68
	AA	89.26 $\pm$ 3.65	86.00 $\pm$ 1.75
	Kappa	83.13 $\pm$ 5.98	74.04 $\pm$ 3.01
Scene-Context Relation Description	OA	87.00 $\pm$ 5.38	76.26 $\pm$ 4.82
	AA	89.47 $\pm$ 2.05	86.27 $\pm$ 1.44
	Kappa	83.37 $\pm$ 6.19	73.33 $\pm$ 5.12
C2TD with Correct Concepts	OA	88.34 $\pm$ 2.09	77.68 $\pm$ 2.14
	AA	89.93 $\pm$ 2.37	86.42 $\pm$ 1.25
	Kappa	84.76 $\pm$ 2.80	74.81 $\pm$ 2.39

class-level descriptions, scene-dependent contextual information, or the mere presence of concept-like tokens.

As shown in Table 9, C2TD with Correct Concepts achieves the best performance on both the PU and IP datasets. For the PU dataset, C2TD with Correct Concepts achieves OA, AA, and Kappa values of 88.34%, 89.93%, and 84.76%, respectively, improving over Class-Name Only by 0.94%, 1.45%, and 2.97%. For the IP dataset, it achieves OA, AA, and Kappa values of 77.68%, 86.42%, and 74.81%, respectively, with corresponding improvements of 0.72%, 0.81%, and 0.67%. These results indicate that introducing concept-level semantic information into the textual branch is more effective than relying only on class names.

More importantly, C2TD with Correct Concepts achieves better overall results than C2TD with Shuffled Concepts on both datasets. Specifically, it improves OA, AA, and Kappa by 1.88%, 0.01%, and 2.06% on PU, and by 1.00%, 0.55%, and 1.13% on IP, respectively. Although the AA gain on PU is marginal, the consistent advantages in OA and Kappa on PU, together with the improvements across all three metrics on IP, indicate that correct class-to-concept matching is beneficial for cross-domain few-shot classification. Since C2TD with Shuffled Concepts uses the same template and concept vocabulary as C2TD with Correct Concepts but destroys the correspondence between class names and concept labels, this comparison directly shows that the effectiveness of C2TD mainly comes from semantically correct concept-level guidance, rather than from longer prompts or additional concept-like tokens.

Compared with Class-Specific Rich Description, C2TD with Correct Concepts improves the OA, AA, and Kappa values by 1.09%, 0.67%, and 1.63% on PU, respectively; the corresponding improvements on IP are 0.67%, 0.42%, and 0.77%. Compared with Scene-Context Relation Description, C2TD with Correct Concepts improves the OA, AA, and Kappa values by 1.34%, 0.46%, and 1.39% on PU, respectively; the corresponding improvements on IP are 1.42%, 0.15%, and 1.48%. These results indicate that structured concept-level semantics provide more effective semantic guidance than simply enriching class descriptions or introducing scene-dependent inter-class relation descriptions. Therefore, the results in Table 9 further verify that C2TD is not merely an advanced prompt-enrichment strategy, but a structured concept-guided text construction method that introduces transferable and interpretable semantic priors for cross-domain visual-textual alignment.

4.7. Compared with other methods

This section presents a quantitative and visual comparison of the classification performance of various supervised and FSL methods, using a limited number of labeled samples from five HSI target domain datasets.

Table 10

Overall Accuracy(%) of classification Accuracy (%) using SVM, 3D-CNN, DFSL, DCFSL, CMFSL, Gia-CFSL, MRLF, HyMuT, SGPL and CGVT-FSL on the HT Dataset, with Five labeled samples per Class.

Class	SVM	3D-CNN	DFSL	DCFSL	CMFSL	Gia-CFSL	MRLF	HyMuT	SGPL	CGVT-FSL
1	81.21	86.62	87.05	88.48	88.39	88.91	88.84	89.55	85.37	87.84
2	45.80	61.36	83.07	86.27	91.47	89.05	89.05	90.56	72.11	90.66
3	98.62	89.99	95.87	95.65	97.34	94.94	97.66	96.10	92.50	98.21
4	88.88	63.91	92.08	95.76	98.26	93.97	97.15	96.25	74.12	95.77
5	74.08	67.04	88.59	88.94	91.18	97.24	91.18	97.90	96.54	98.68
6	82.30	53.39	82.34	87.66	84.25	81.47	81.31	81.31	87.18	90.25
7	12.87	40.98	76.42	76.10	89.50	73.75	83.58	78.61	78.61	85.27
8	14.03	46.79	52.44	58.19	54.78	49.63	52.75	51.49	72.33	49.41
9	75.92	43.86	76.68	77.73	72.65	65.12	75.90	69.93	79.61	81.52
10	5.29	49.89	61.54	52.40	63.24	68.63	68.38	71.35	73.46	75.32
11	28.65	31.20	63.58	76.93	74.96	65.46	76.32	65.46	81.97	78.50
12	5.39	43.92	57.67	38.88	59.36	65.92	62.77	62.44	74.39	63.98
13	10.35	35.53	89.05	80.69	93.84	90.13	90.62	90.13	37.01	88.58
14	95.01	58.83	82.86	99.08	82.86	91.96	92.32	93.50	65.30	91.96
15	92.83	62.63	90.63	98.17	98.84	99.63	98.12	98.12	85.54	99.51
OA	45.55 ± 3.11	54.60 ± 3.82	76.61 ± 3.10	79.61 ± 1.79	81.25 ± 1.87	78.39 ± 1.72	79.79 ± 1.77	80.66 ± 3.11	74.93 ± 1.02	83.35 ± 2.49
AA	54.08 ± 2.55	55.73 ± 3.06	78.66 ± 2.74	82.60 ± 2.70	83.44 ± 1.65	80.53 ± 1.47	81.79 ± 1.58	82.60 ± 2.70	77.07 ± 1.26	85.55 ± 2.06
Kappa	44.80 ± 3.27	51.03 ± 3.42	74.75 ± 3.33	77.97 ± 2.02	79.73 ± 2.02	76.66 ± 1.85	79.32 ± 1.91	79.10 ± 3.36	73.00 ± 1.09	82.03 ± 2.67

Table 11

Overall Accuracy (%) of classification Accuracy (%) using SVM, 3D-CNN, DFSL, DCFSL, CMFSL, Gia-CFSL, MRLF, HyMuT, SGPL and CGVT-FSL on the IP Dataset, with Five labeled samples per Class.

Class	SVM	3D-CNN	DFSL	DCFSL	CMFSL	Gia-CFSL	MRLF	HyMuT	SGPL	CGVT-FSL
1	72.20	95.12	90.73	92.20	95.37	90.73	96.34	96.34	30.42	99.51
2	34.27	37.70	34.24	39.52	49.11	44.05	48.51	53.01	75.19	65.04
3	39.18	19.77	39.71	49.72	52.95	49.59	55.72	61.38	48.61	64.46
4	50.34	32.51	58.19	80.22	77.20	79.09	86.94	83.84	45.97	93.97
5	69.75	88.45	66.46	73.31	72.11	72.47	78.95	77.22	81.76	80.94
6	66.36	73.65	74.36	84.92	84.30	83.50	93.01	86.03	90.50	96.94
7	89.13	81.82	93.48	96.96	99.57	96.52	99.13	100.00	47.27	100.00
8	68.73	53.35	74.42	82.24	90.97	87.08	84.08	90.63	78.72	96.70
9	86.67	100.00	98.00	100.00	99.33	98.67	98.67	99.33	16.41	100.00
10	37.49	41.35	48.78	58.67	61.39	59.02	67.79	66.23	47.29	73.39
11	33.96	66.71	56.92	58.70	51.78	62.65	66.16	59.22	78.72	71.45
12	31.43	37.40	37.53	46.26	52.04	45.80	53.03	53.08	47.66	59.80
13	86.50	85.71	94.60	97.85	99.70	98.90	99.65	98.20	57.87	99.50
14	62.93	62.57	77.58	85.86	80.20	78.60	87.83	81.44	79.37	91.90
15	28.08	56.42	66.01	74.17	83.52	69.37	72.70	78.64	49.58	89.55
16	90.91	90.36	96.82	99.55	99.43	98.64	99.09	99.09	42.05	99.55
OA	45.85 ± 2.44	54.76 ± 0.03	57.26 ± 2.38	64.06 ± 3.54	64.53 ± 3.07	64.62 ± 3.78	70.18 ± 2.05	68.51 ± 3.49	64.68 ± 0.95	77.68 ± 2.14
AA	59.24 ± 1.36	63.93 ± 0.02	69.24 ± 1.93	76.26 ± 1.96	78.06 ± 1.16	75.92 ± 2.24	80.47 ± 1.89	80.23 ± 1.76	58.10 ± 0.89	86.42 ± 1.25
Kappa	39.68 ± 2.48	48.72 ± 0.03	51.97 ± 2.39	59.58 ± 3.83	60.24 ± 3.03	60.04 ± 3.97	66.36 ± 2.39	64.56 ± 3.69	60.61 ± 0.94	74.81 ± 2.39

1) Statistical Table of Classification Results

Tables 10–14 present the OA, AA, Kappa, and per-class precision for cross-domain HSI classification. For all methods, five labeled samples per class are randomly selected as supervised samples in the target domain. To mitigate the impact of random sampling, all experiments are repeated 10 times. As shown in Table 10, CGVT-FSL achieves an OA of 83.35% and an AA of 85.55% on the HT dataset, outperforming other methods including SVM, 3D-CNN, DFSL, DCFSL, CMFSL, Gia-CFSL, MRLF, SGPL and HyMuT. These methods achieve OAs ranging from 45.55% to 81.25% and AAs ranging from 54.08% to 83.44%. This comparison reveals that FSL methods outperform supervised methods, emphasizing the effectiveness of cross-domain episodic training mechanisms. Notably, CGVT-FSL achieves the highest accuracy, demonstrating the superiority of the VT-FSL strategy in cross-domain HSI classification.

Similarly, we perform 5-shot classification on the IP dataset to evaluate the performance of the ten different methods. As shown in Table 11, spatial-spectral feature extraction methods such as 3D-CNN methods outperform the SVM method with few labeled samples. Moreover, DCFSL, CMFSL, Gia-CFSL, MRLF, HyMuT, SGPL and CGVT-FSL all achieve promising results, further confirming the effectiveness of cross-domain episodic training mechanisms. Particularly, the proposed

CGVT-FSL exploits concept-guided textual semantics to enhance domain invariance, achieving the best classification accuracy on the IP dataset.

Tables 12–14 present the 5-shot classification accuracy of the ten different methods on the PU, HC and HH datasets, respectively. From the results displayed in these tables, it is evident that FSL methods consistently perform strongly even with limited labeled samples. Notably, CGVT-FSL surpasses other FSL methods across all datasets, highlighting the effectiveness of the proposed VTDFormer in extracting domain-invariant representations and improving cross-domain HSI classification performance.

As mentioned above, compared with other advanced FSL methods, CGVT-FSL achieves the fewest misclassified samples in cross-domain HSI classification. The following factors contribute to its superior performance:

- Even in datasets (e.g., PU) with strong linear separability, traditional machine learning methods (e.g., SVM) can perform comparably to certain deep learning methods (e.g., 3D-CNN). However, due to the large-scale and high-dimensional nature of HSI data, deep learning methods generally yield better results.

Table 12

Overall Accuracy(%) of classification Accuracy (%) using SVM, 3D-CNN, DFSL, DCFSL, CMFSL, Gia-CFSL, MRLF, HyMuT, SGPL and CGVT-FSL on the PU Dataset, with Five labeled samples per Class.

Class	SVM	3D-CNN	DFSL	DCFSL	CMFSL	Gia-CFSL	MRLF	HyMuT	SGPL	CGVT-FSL
1	88.98	59.82	76.38	74.97	77.35	77.27	75.70	81.58	88.82	86.91
2	83.91	63.05	78.22	84.87	82.20	86.40	79.22	84.22	95.07	89.47
3	39.98	68.91	51.89	64.84	63.95	60.92	65.16	69.96	61.16	80.53
4	60.22	77.31	91.05	94.45	92.38	92.17	89.60	90.26	79.30	94.59
5	95.44	90.77	99.19	99.44	99.49	99.02	99.77	98.20	98.06	99.97
6	37.12	63.40	60.58	76.02	84.90	72.93	70.75	69.76	69.57	81.04
7	40.62	87.64	69.80	80.01	81.74	77.14	90.95	79.24	49.86	94.78
8	68.17	57.27	64.70	65.43	73.14	71.86	83.77	74.23	90.67	85.86
9	99.13	95.57	98.66	99.16	99.01	98.77	98.60	97.34	69.39	96.24
OA	64.12 ± 4.55	65.74 ± 1.77	75.17 ± 2.95	80.94 ± 4.30	81.72 ± 3.98	81.70 ± 2.58	79.56 ± 4.93	81.58 ± 4.02	82.98 ± 3.59	88.34 ± 2.09
AA	68.18 ± 2.27	73.72 ± 1.01	76.72 ± 2.10	82.13 ± 2.26	83.79 ± 1.56	81.83 ± 1.52	83.72 ± 2.31	82.77 ± 1.68	77.99 ± 2.81	89.93 ± 2.37
Kappa	55.59 ± 4.64	57.37 ± 1.97	68.04 ± 3.70	75.44 ± 4.85	76.64 ± 4.60	76.25 ± 3.04	73.89 ± 5.48	76.15 ± 4.58	77.95 ± 4.12	84.76 ± 2.80

Table 13

Overall Accuracy(%) of classification Accuracy (%) using SVM, 3D-CNN, DFSL, DCFSL, CMFSL, Gia-CFSL, MRLF, HyMuT, SGPL and CGVT-FSL on the HC Dataset, with Five labeled samples per Class.

Class	SVM	3D-CNN	DFSL	DCFSL	CMFSL	Gia-CFSL	MRLF	HyMuT	SGPL	CGVT-FSL
1	28.33	24.54	70.58	72.59	65.16	67.97	73.22	73.30	82.19	79.08
2	25.57	18.85	47.83	46.22	60.11	43.05	59.15	52.38	65.94	57.18
3	10.02	16.23	62.03	60.39	74.24	62.09	70.36	76.94	50.65	80.68
4	11.13	29.30	74.48	90.50	95.68	86.41	92.33	94.49	47.02	97.33
5	12.26	12.53	89.77	89.09	93.60	86.23	95.26	92.51	24.34	97.54
6	40.34	9.89	23.80	24.27	41.48	34.93	38.40	44.15	32.65	56.19
7	10.59	46.17	82.19	83.36	85.86	76.80	83.38	78.93	45.36	87.44
8	52.22	35.49	34.60	35.50	47.82	41.45	33.57	44.32	66.55	55.62
9	4.97	6.03	24.72	33.15	43.83	30.06	36.56	41.80	62.96	55.54
10	26.61	40.67	57.09	62.84	79.33	69.18	76.46	80.03	34.56	86.95
11	36.39	61.64	65.18	68.52	73.24	61.17	62.96	75.50	69.23	77.05
12	11.07	14.64	55.90	54.36	62.27	50.48	57.20	66.10	39.03	76.17
13	52.26	11.38	38.05	42.63	41.15	38.59	40.07	47.56	35.03	54.26
14	65.55	46.47	52.34	58.51	49.72	58.56	52.43	58.62	73.69	63.17
15	19.44	13.00	77.58	66.00	80.67	67.76	82.42	69.15	74.94	81.27
16	77.55	81.39	81.92	87.64	89.47	87.89	91.06	91.33	91.71	93.40
OA	34.87 ± 3.24	41.76 ± 4.03	63.43 ± 4.13	66.97 ± 4.69	64.53 ± 3.07	70.20 ± 3.29	69.70 ± 4.73	72.18 ± 2.82	67.35 ± 0.61	77.29 ± 3.00
AA	30.27 ± 2.46	29.26 ± 1.42	58.63 ± 2.09	60.97 ± 3.20	67.73 ± 2.50	60.16 ± 3.69	65.30 ± 3.18	67.95 ± 2.01	55.99 ± 1.10	74.93 ± 2.48
Kappa	27.23 ± 5.61	34.08 ± 3.23	58.28 ± 4.35	62.12 ± 5.15	65.89 ± 3.55	60.90 ± 5.69	65.22 ± 5.11	68.01 ± 3.09	62.18 ± 0.61	73.81 ± 3.30

Table 14

Overall Accuracy(%) of classification Accuracy (%) using SVM, 3D-CNN, DFSL, DCFSL, CMFSL, Gia-CFSL, MRLF, HyMuT, SGPL and CGVT-FSL on the HH Dataset, with Five labeled samples per Class.

Class	SVM	3D-CNN	DFSL	DCFSL	CMFSL	Gia-CFSL	MRLF	HyMuT	SGPL	CGVT-FSL
1	52.98	48.13	75.42	80.06	71.92	77.76	83.74	81.30	87.95	88.98
2	38.88	40.37	68.44	76.82	67.77	68.18	76.07	72.83	60.13	81.76
3	80.32	80.08	87.53	91.80	79.44	88.96	87.14	90.24	84.40	89.71
4	53.91	53.32	81.01	86.28	80.14	85.27	87.41	86.74	99.06	92.80
5	10.66	6.67	61.36	70.70	76.49	65.63	69.48	72.01	43.52	79.89
6	39.36	45.80	80.80	86.89	82.02	82.86	87.02	87.57	90.52	90.35
7	30.46	30.29	42.57	56.77	52.36	52.67	60.64	58.39	65.26	70.66
8	6.43	4.29	29.03	36.37	48.93	26.78	33.80	40.95	41.93	53.73
9	34.92	37.15	87.32	93.60	86.87	93.36	93.92	94.26	93.95	97.65
10	12.28	7.21	55.91	59.46	54.85	56.92	60.48	62.54	74.23	74.70
11	9.11	13.03	48.05	51.72	59.73	53.73	56.09	58.48	47.40	65.59
12	9.06	9.03	59.92	60.99	44.55	60.37	59.18	62.29	61.58	69.37
13	16.17	24.24	48.74	45.97	47.36	51.48	50.47	51.86	85.65	60.98
14	23.83	43.91	68.00	76.60	60.86	72.70	77.71	77.55	39.41	86.62
15	3.04	5.15	81.89	88.00	93.89	84.95	86.21	86.95	12.11	96.95
16	51.65	48.56	71.37	60.82	82.12	67.39	73.29	70.50	94.94	91.41
17	29.82	29.78	72.40	79.76	80.98	78.85	82.80	76.32	90.42	87.91
18	7.32	4.73	63.16	81.14	91.39	77.97	82.12	74.84	36.96	95.38
OA	38.85 ± 8.60	40.20 ± 3.96	72.85 ± 4.51	78.06 ± 3.01	73.36 ± 4.07	76.63 ± 3.01	79.34 ± 2.53	79.28 ± 3.09	82.88 ± 1.17	86.06 ± 2.16
AA	28.34 ± 0.97	29.54 ± 0.93	65.72 ± 2.51	71.32 ± 1.96	70.09 ± 2.80	69.21 ± 2.81	72.64 ± 2.50	72.53 ± 2.66	67.19 ± 0.73	81.91 ± 2.30
Kappa	31.31 ± 6.76	32.53 ± 3.02	66.49 ± 4.84	72.55 ± 3.41	67.10 ± 4.38	70.82 ± 3.47	74.02 ± 2.86	74.05 ± 3.56	78.27 ± 1.36	82.25 ± 2.56

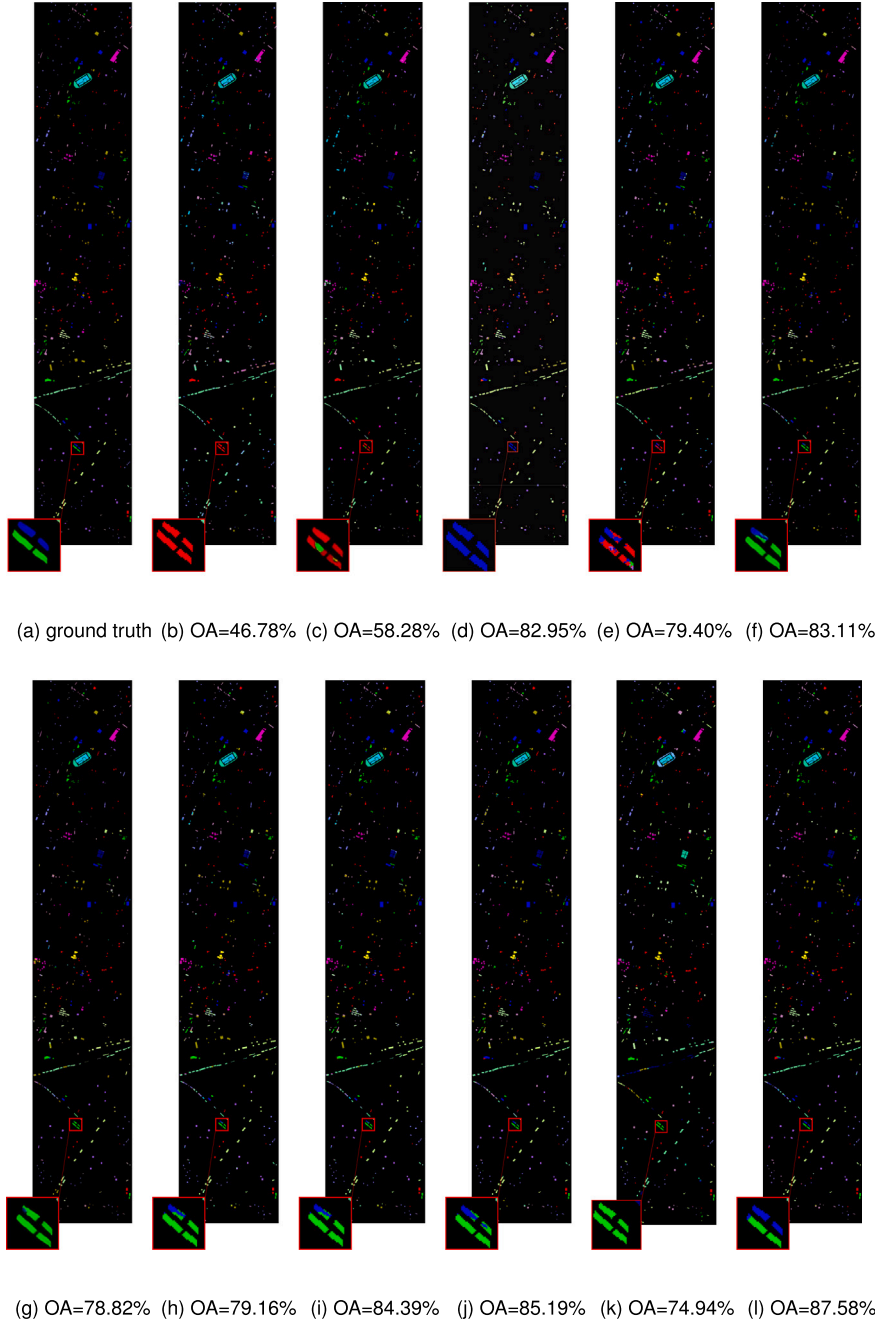


Fig. 8. HT dataset: classification map using (a) ground truth, (b) SVM, (c) 3D-CNN, (d) SSRN, (e) DFSL, (f) DCFSL, (g) CMFSL, (h) Gia-CFSL, (i) MRLF, (j) HyMuT, (k) SGPL and (l) CGVT-FSL.

- Traditional machine learning and deep learning methods (e.g., SVM and 3D-CNN) fail to learn transferable knowledge from the source domain, hindering their ability to generalize effectively to the target domain. As a result, their generalization performance is limited when only a few labeled samples are available.
- Previous FSL methods (e.g., DFSL, DCFSL, and Gia-CFSL) primarily focus on domain-level feature alignment while neglecting inter-class semantic relationships. In contrast, CGVT-FSL introduces the C2TD module, which generates concept-embedded textual descriptions that capture both intra-class consistency and inter-class distinctiveness, enabling more discriminative semantic alignment.
- Current FSL methods (e.g., CMFSL, MRLF, and HyMuT) mainly rely on visual features and underutilize linguistic information. Although

SGPL adopts semantic information, it uses simple class-label descriptions with limited semantic richness. In contrast, CGVT-FSL introduces the C2TD method to inject transferable concept knowledge into text representations, and further integrates VTDFormer and SSTDAM for visual-linguistic alignment and dual-domain adversarial learning, thereby reducing modality and domain gaps and improving cross-domain generalization.

2) Classification Maps of Different Methods

To demonstrate the effectiveness of our proposed CGVT-FSL framework, we visually compare its classification performance with that of both supervised and FSL methods. Fig. 8 presents classification maps generated for the HT dataset using ten different methods, each trained

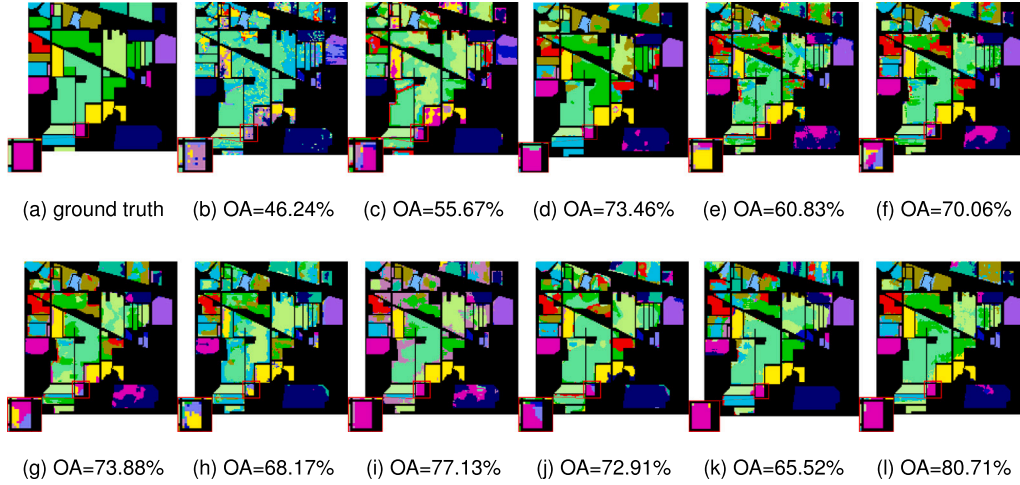


Fig. 9. IP dataset: classification map using (a) ground truth, (b) SVM, (c) 3D-CNN, (d) SSRN, (e) DFSL, (f) DCFSL, (g) CMFSL, (h) Gia-CFSL, (i) MRLF, (j) HyMuT, (k) SGPL and (l) CGVT-FSL.

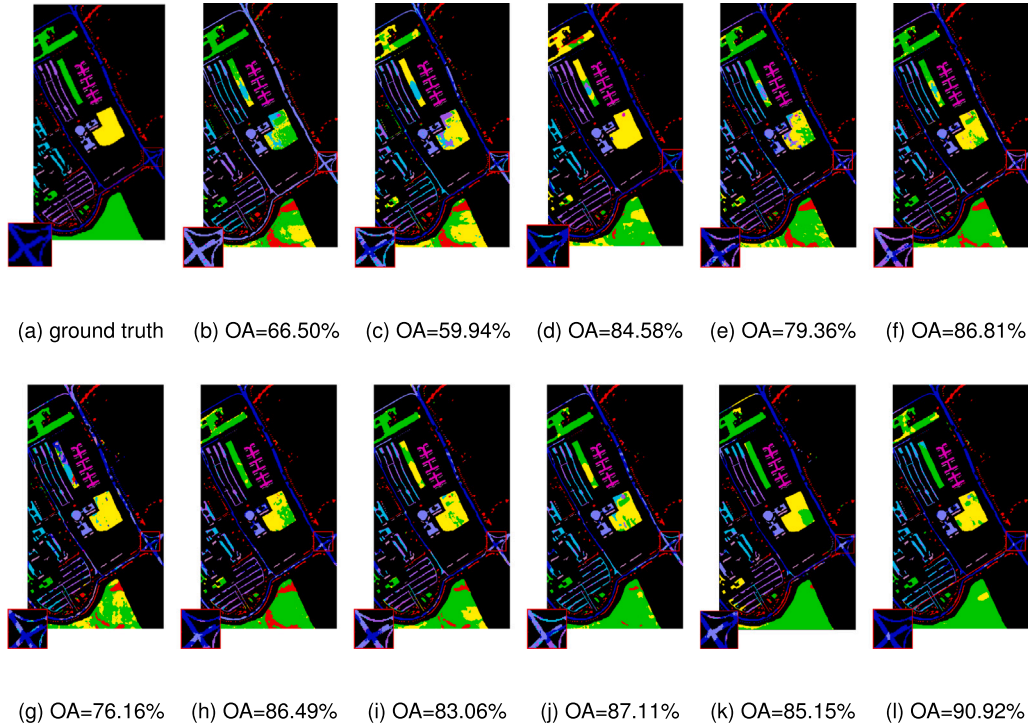


Fig. 10. PU dataset: classification map using (a) ground truth, (b) SVM, (c) 3D-CNN, (d) SSRN, (e) DFSL, (f) DCFSL, (g) CMFSL, (h) Gia-CFSL, (i) MRLF, (j) HyMuT, (k) SGPL and (l) CGVT-FSL.

with five labeled samples per class. Visual analysis reveals that supervised methods tend to assign more query samples to incorrect classes compared to FSL methods. Notably, the classification map generated by CGVT-FSL is closely aligned with the ground-truth map, highlighting that the integration of VTDFormer and SSTDAM enables robust visual-linguistic alignment and effectively improves cross-domain HSI classification. Similarly, Fig. 9 displays classification maps obtained for the IP dataset using the same ten methods. Consistent with the HT dataset, the classification map generated by CGVT-FSL demonstrates superior alignment with the ground-truth map, further validating its efficacy in cross-domain HSI classification.

Furthermore, Figs. 10, 12 and 13 present classification maps for the PU, HC and HH datasets, respectively, with each method trained using

five labeled samples. A comparison of classification maps across different methods reveals that those produced by CGVT-FSL consistently align most closely with the ground truth. This further demonstrates that the spatial-spectral features extracted by CGVT-FSL are more robust and discriminative than those from competing methods in cross-domain HSI classification.

4.8. Influence of the labeled sample size

Moreover, we further evaluate the robustness of CGVT-FSL under varying numbers of labeled samples. Specifically, experiments are conducted with 1, 2, 3, 4, and 5 labeled samples per class from the target domain for few-shot learning methods. For supervised approaches that do not follow an episodic training paradigm, the same numbers of

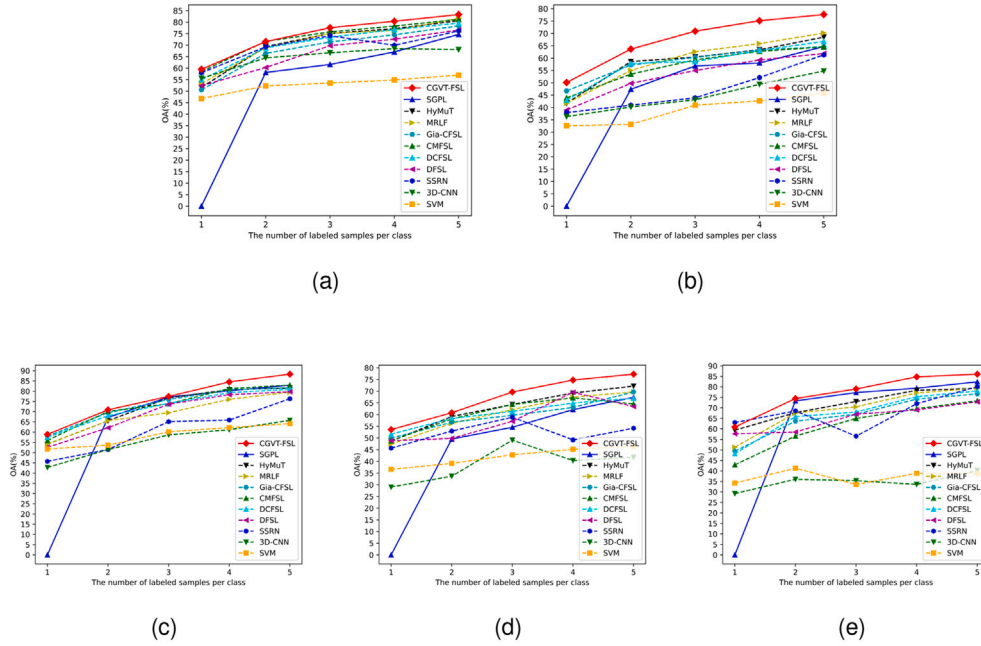


Fig. 11. Influence of labeled sample size on the five target domain datasets: (a) HT (b) IP (c) PU (d) HC (e) HH datasets.

labeled samples per class are randomly selected for training. As illustrated in Fig. 11, the OA of all methods generally improves as the number of labeled samples increases. Notably, CGVT-FSL consistently achieves superior performance across different sample sizes, indicating that the incorporation of domain-invariant class-label text representations and concept-level semantic embeddings effectively enhances the model's adaptability to varying levels of labeled data availability.

In addition, since SGPL adopts a pre-training strategy on the source domain followed by fine-tuning on the target domain under the few-shot setting, it cannot be directly applied in the 1-shot scenario. Specifically, the fine-tuning stage requires at least two labeled samples per class (for constructing both support and query sets), whereas only a single labeled sample is available in the 1-shot case. Consequently, the complete SGPL framework is infeasible under the 1-shot setting, and thus no experimental results are reported for SGPL in this condition.

4.9. Computational complexity

Additionally, we assess the computational cost and classification performance of six comparative methods together with CGVT-FSL on the HC and HH datasets. As reported in Table 15, the training time, inference time, and OA of all compared methods are presented separately. The results show that CGVT-FSL achieves competitive inference efficiency while delivering the best classification performance on both datasets. Specifically, on the HC dataset, CGVT-FSL achieves an OA of 77.29% with an inference time of 8.09 s, which is close to that of DCFSL (7.92 s) and lower than those of Gia-CFSL (8.72 s), MRLF (12.53 s), and HyMuT (10.07 s). On the HH dataset, CGVT-FSL attains an OA of 86.06% with an inference time of 7.99 s, which is also close to DCFSL (7.68 s) and HyMuT (7.79 s), and lower than Gia-CFSL (8.22 s) and MRLF (10.54 s). Although CGVT-FSL does not have the shortest training time, its training cost remains within a reasonable range among the compared methods, while its inference time stays competitive. Overall, these results indicate that CGVT-FSL achieves a favorable balance between computational cost and classification performance, making it suitable for large-scale cross-domain HSI analysis where both predictive accuracy and inference efficiency are important.

4.10. Reliability analysis of class-to-concept label matching

Following recent recommendations on transparent human evaluation of large language models [45], we do not assume that the class-to-concept label assignments produced by the LLM in C2TD are fully reliable. Instead, we explicitly assess their reliability through a quantitative audit and use the results to guide subsequent manual review.

Specifically, we first conduct a manual-review-based audit of the class-to-concept label matching results to quantify their reliability and characterize potential mismatch patterns. The audit is performed at the class level under the predefined C2TD taxonomy across the same three concept dimensions used in C2TD, namely object type, density, and specific heat capacity. For each land-cover class, the LLM-generated concept labels are examined by authors with remote-sensing expertise according to fixed class-level criteria derived from (i) the semantic meaning of the class name, (ii) general remote-sensing knowledge of land-cover properties, and (iii) the examples and figure annotations provided in this manuscript, with priority given to evidence directly relevant to the target class. The reviewed labels are then used as reference annotations for quantitative evaluation.

Following the statistical discussion in [60], simple Matching Accuracy alone is insufficient for evaluating agreement between two label sources on nominal categorical variables, as it does not account for agreement that may occur purely by chance. Therefore, in addition to the direct matching rate, we report a chance-corrected agreement statistic, denoted as K_{agr} , to measure the consistency between the LLM-generated labels and the manually reviewed reference labels. The statistic is defined as:

$$K_{agr} = \frac{P_o - P_e}{1 - P_e}, \quad (32)$$

where P_o denotes the observed agreement (i.e., Matching Accuracy), and P_e denotes the expected agreement by chance, estimated from the marginal label distributions of the two sources. In this setting, Matching Accuracy reflects the direct matching rate, while K_{agr} provides a more reliable estimate of agreement beyond chance.

The audit is conducted across three representative LLM backends, namely DeepSeek [61], Doubao [62], and Hunyuan [63]. As shown in

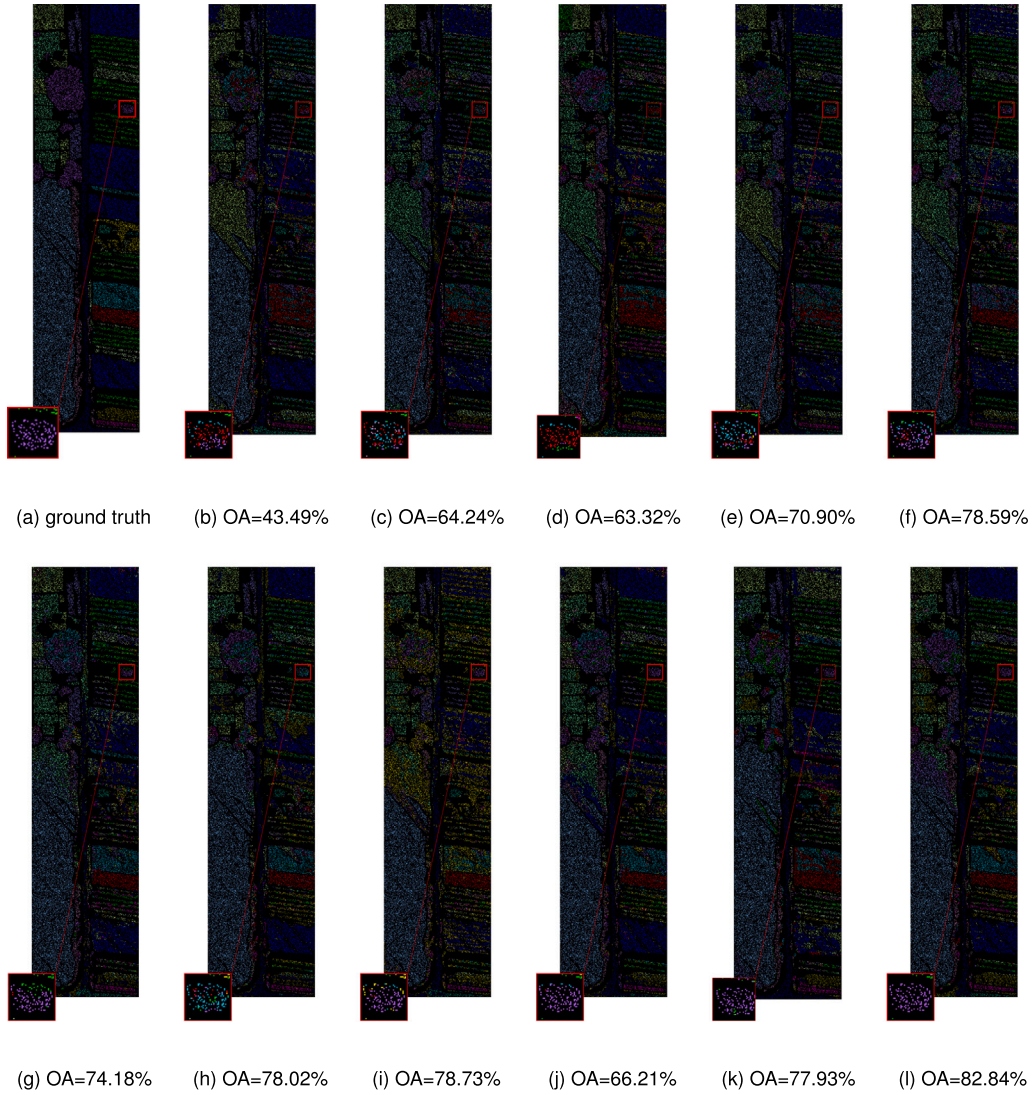


Fig. 12. HC dataset: classification map using (a) ground truth, (b) SVM, (c) 3D-CNN, (d) SSRN, (e) DFSL, (f) DCFSL, (g) CMFSL, (h) Gia-CFSL, (i) MRLF, (j) HyMuT, (k) SGPL and (l) CGVT-FSL.

Table 16, the reliability of class-to-concept label matching is clearly dimension-dependent rather than uniformly consistent across backends. Object type achieves consistently high agreement across all three LLMs, indicating that this dimension is relatively stable and reliable. In contrast, density exhibits the largest variation across different backends, suggesting higher sensitivity to model-specific behavior. The agreement for specific heat capacity is also lower than that for object type, indicating that fine-grained physical-property assignment is more uncertain than coarse semantic categorization.

Importantly, these findings directly motivate the necessity of manual review. If the LLM-generated labels were uniformly reliable across all concept dimensions, manual correction would be unnecessary. However, the dimension-dependent discrepancies observed in Table 16 indicate that certain dimensions—particularly density and specific heat capacity—require careful human verification and correction. The audit further reveals three representative error tendencies, which provide guidance for the manual review process. The first is *scene-context*

neglect, observed in classes such as bare soil-park and bare soil-school, where the LLM overemphasizes material cues while underweighting contextual information, indicating that contextual semantics should be explicitly verified. The second is *density underestimation*, where some vegetation-related classes are assigned to lower density levels than expected, suggesting that density-related attributes should be carefully re-evaluated using domain knowledge. The third is *middle-category collapse* in the specific-heat-capacity dimension, where uncertain samples tend to be mapped to intermediate categories (e.g., “medium”) rather than being properly differentiated, indicating that reviewers should avoid default intermediate assignments and instead enforce more physically grounded and discriminative labeling.

Overall, the quantitative audit Table 16 and the subsequent error-pattern analysis jointly establish a structured mechanism for identifying, interpreting, and correcting LLM-induced biases, thereby improving the reliability and interpretability of the concept construction process.

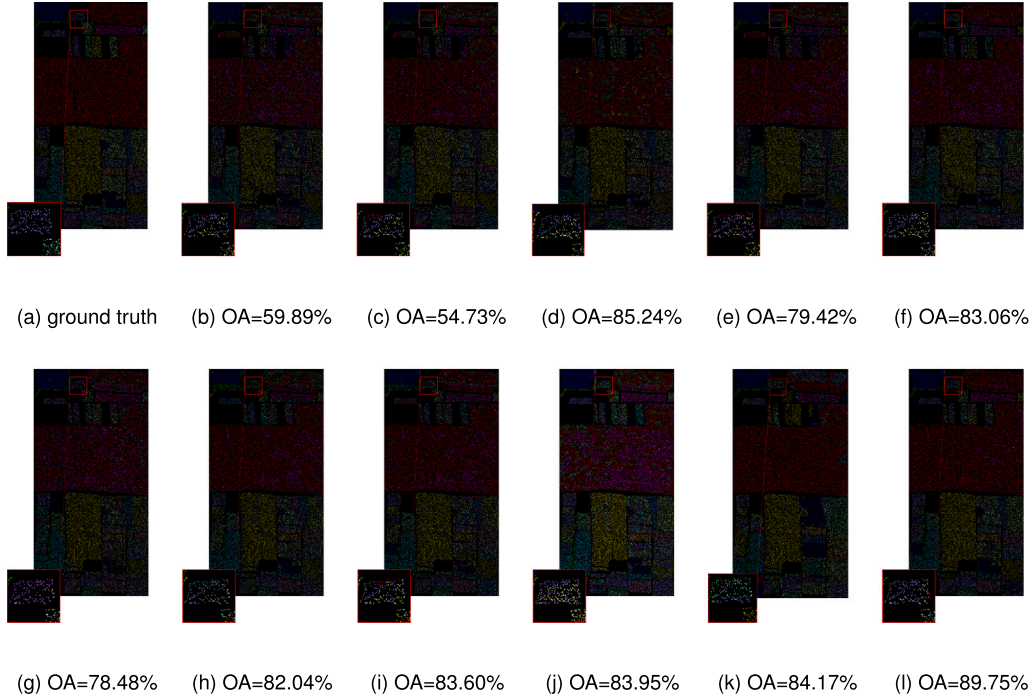


Fig. 13. HH dataset: classification map using (a) ground truth, (b) SVM, (c) 3D-CNN, (d) SSRN, (e) DFSL, (f) DCFSL, (g) CMFSL, (h) Gia-CFSL, (i) MRLF, (j) HyMuT, (k) SGPL and (l) CGVT-FSL.

Table 15

Time and OA using six different methods on the HC and HH Datasets.

Dataset	Metric	DCFSL	CMFSL	Gia-CFSL	MRLF	HyMuT	CGVT-FSL
HC	Training Time	1162.86 s	952.51 s	2865.39 s	1984.26 s	3380.53 s	2053.28 s
	Inference Time	7.92 s	6.28 s	8.72 s	12.53 s	10.07 s	8.09 s
	OA	66.97%	64.53%	70.20%	69.70%	72.18%	77.29%
HH	Training Time	1407.35 s	960.83 s	3037.36 s	1825.49 s	3878.34 s	1846.53 s
	Inference Time	7.68 s	6.16 s	8.22 s	10.54 s	7.79 s	7.99 s
	OA	78.06%	73.36%	76.63%	79.34%	79.28%	86.06%

Table 16

Quantitative audit of LLM-generated class-to-concept label assignments.

Version	Object type Accuracy	Object type Agreement	Density Accuracy	Density Agreement	Heat-capability Accuracy	Heat-capability Agreement
DeepSeek	97.85%	0.968	88.17%	0.788	60.22%	0.266
Doubao	95.70%	0.935	38.71%	0.182	72.04%	0.496
Hunyuan	95.70%	0.935	69.89%	0.437	77.42%	0.588

5. Conclusion

This work proposes CGVT-FSL, a novel VT-FSL framework that bridges the gap between visual and linguistic modalities for cross-domain HSI classification. Specifically, we first introduce the C2TD method, which embeds transferable concept-level semantics into class-label text descriptions. This enables more interpretable and domain-agnostic textual representations that capture both intra-class and inter-class relationships. Next, we propose the VTDFormer backbone, which extracts both visual and textual spatial-spectral features across domains, facilitating visual-linguistic-prototype alignment to learn domain-invariant feature representations. Additionally, we design the SSTDAM module to reduce domain shift by performing joint source-target domain and visual-linguistic domain adversarial learning. In real-world applications, CGVT-FSL effectively generalizes to the target domain by leveraging concept-embedded class-label text and a small

number of labeled samples per class, significantly reducing the cost of labeling. Extensive experiments across six HSI datasets demonstrate the superior performance of CGVT-FSL compared to existing state-of-the-art methods. However, a prominent limitation remains: high computational complexity. Future work will focus on exploring lightweight architectures to enhance efficiency.

CRedit authorship contribution statement

Haojin Tang: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Jiaqing Peng:** Writing – review & editing, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Hongyi Li:** Writing – original draft, Software, Methodology. **Xiaofei Yang:** Writing – review & editing, Methodology, Funding acquisition. **Dong Tang:**

Supervision, Project administration. **Xin Lin:** Writing – review & editing, Investigation. **Weixin Xie:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my code in the manuscript.

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